

Comparing the Effectiveness of Traditional and Machine Learning
Methods in Stock Prediction

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Abstract

Background: Financial markets are influenced by a variety of factors, and stock prices are an important indicator of economic trends and overall economic conditions. Developments in data analytics have fuelled quantitative research on stock market forecasting. This dissertation will compare the performance of traditional statistical methods and machine learning methods in stock prediction.

Aim: The aim of this thesis is to deepen the understanding of stock price prediction by finding the most effective predictive models in linear regression, random forest and Arima through a data-driven approach.

Methods: This dissertation will collect closing prices from January 2020 to December 2022 for Nasdaq Composite, Tesla, Amazon, Netflix and Google. Then use Linear Regression, Random Forest and Arima Model to study and predict the stock prices and finally compare the performance of different models.

Results: The results of this research show that the Nasdaq Composite has a higher prediction accuracy than a specific stock when it comes to stock prediction. For model comparison, Random Forest performs best in predicting long-term stock price trends, with accuracy above 90% in all five datasets, and Linear Regression has the highest accuracy in predicting short-term stock prices. In addition, the study also found the relationship between macroeconomic variables and stock prices.

Conclusion: This study proves the predictability of the stock market and investigates the predictive performance of relevant forecasting models, providing a contribution to the quantitative study of financial markets. However, the inherent volatility and complexity of financial markets still leave many limitations to forecasting research. Further research needs to focus on the characteristics of new variables such as social media sentiment and use more advanced methods for forecasting.

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Chapter 1. Introduction

1.1 Background

The first stock exchange in the world was the London Stock Exchange, founded in 1773. Initially, traders exchanged shares in a café (Chen, 2023). With the development of the market economy, stocks have demonstrated their unique appeal and they are pivotal to the global economy. Stock markets can provide a platform for companies to raise capital through public offerings (IPO), in which companies issue shares to the public to raise funds to expand their companies or innovate their products or services. For individuals, the stock market provides them with the opportunity to participate in the operations of a company and invest for profit. In addition, stock market and national economic development are closely related (Kamley et al., 2013). If the stock market rises, then the country's economic growth will be high. If the stock market falls, then the country's economic growth will fall (Fama, 1965). For example, when the stock market is buoyant, consumers and businesses have more money and expectations, which leads to more spending and higher GDP (Hall, 2022). Therefore, the stock market is important for economic activities and affects everyone's life.

Since the beginning of 2020, Tesla's share price has shown phenomenal growth, it has soared from \$28 to \$380 in just two years, capturing the attention of the world (Liu, 2021). The American electric car manufacturing giant is capturing the global market at an alarming rate due to the global shift towards renewable energy sources and the gradual replacement of conventional cars by electric cars, a thesis well supported by the performance of the stock market (Capital.com, 2021). Therefore, the stock market is a good source of information about industry developments, and this study will focus on the close price of Nasdaq Composite and four industry leaders - Tesla, Google, Netflix, and Amazon - in the hope of predicting their stock price by Linear Regression, Random Forest and Arima.

No one can deny that stock markets are highly volatile and are subject to many uncertainties and interrelated economic and political factors at the local or global level.

In addition, stock information is often incomplete, noisy, and random, making it difficult for investors and economists to understand the patterns of the stock market and to predict the future (Agrawal et al., 2013). In the past, the public analysed stocks based on fundamentals such as financial condition, industry indicators, macroeconomic factors, and financial news to make purchase decisions (Upadhyay et al., 2012). As the level of investing and trading increases, people look for tools and methods that can increase returns while minimising risk. Technical analysis is a useful tool for predicting stock market behaviour, helping investors to act with greater certainty and to consider the risk and volatility of their investments and to know when it is the best time to buy or sell stocks (Boone, 1998).

Stock forecasting methods at the technical level have gone through different stages. Traditional time series methods include the auto-regressive moving conditional heteroscedasticity model (Arch), auto-regressive moving average (Arima), moving average, exponential smoothing and exponential smoothing and naïve method (Wei et al., 2011). However, based on the nature of financial data, these methods perform poorly in accurately predicting stock market indices. With the development of artificial intelligence, machine learning techniques have been more successful in predicting various fields such as economics, policy, and engineering (Gharehchopogh et al., 2012). Unlike traditional time series methods, these techniques can handle nonlinear, chaotic, noisy, and complex data from stock markets to produce more efficient forecasts (Chen & Hao, 2017). Therefore, their application is attractive for financial market forecasting (Kumbure et al., 2022).

1.2 Research Aim and Objectives

Based on the above discussion, nowadays there are many studies on time series methods, machine learning methods and even deep learning methods for stock market prediction. But a comparison between traditional statistical methods and machine learning methods is lacking.

This research aims to compare the performance of time series methods and machine learning methods. It is hoped to find a better method that can predict stock prices more accurately and reduce prediction errors. It will provide a reference for investors' investment decisions and contribute to the financial field.

The research objectives include:

- i. To review the literatures on Arima, Multiple Regression and Random Forest to stock prediction.
- ii. To investigate the relationship between stock closing price and opening price, high price, low price, volume, interest rate, inflation rate and unemployment rate.
- iii. To develop and validate stock price prediction model.
- iv. To compare the performance of Arima and two machine learning models Multiple Regression and Random Forest.

1.3 Research Questions

1. What is the correlation between the seven predictor variables and the Close price?
2. Which stock price prediction model has the best performance?

1.4 Structure

This dissertation is divided into five parts.

Chapter 1 is introduction. It focuses on the background description of Stock Market and its related forecasting methods. It also describes the research aims and objectives of this study.

Chapter 2 is literature review. It presents the literature structure for this research and reviews the literature on stock forecasting, focusing on the literature review of three models, multiple regression, Arima model and random forest.

Chapter 3 is methodology. It describes the data collection method, research design, data preprocessing and an overview of the three forecasting models for this study.

Chapter 4 is findings and discussion. It describes the prediction results of the three prediction models and compares their performance. Meanwhile, relevant research questions will be discussed and summarised in this section.

Chapter 5 is conclusion. It will summarise the findings of this study in detail and propose limitations for future research.

Chapter 2. Literature Review

This chapter presents theoretical knowledge on the use of macroeconomic variables and historical stock data to predict the return on stocks. It also summarises the literature and theory related to stock forecasting methods such as multiple regression, random forest and Arima models. It contributes to the field of research by helping to develop the research base and revealing some of the research gaps.

2.1 Literature Review Strategy

This dissertation mainly used search engines such as google scholar, Web of Science and Star plus to find relevant literatures. Mainly search keywords such as Stock Market Prediction, Multiple Regression, Arima Model, Random Forest, Machine Learning in Stock Market, Time Series, etc. to search. There are a total of 28,000 search results, and finally I selected 65 literatures as references to explore the difference between time series and machine learning algorithms for stock return prediction.

2.2 The Stock Market Prediction

Stock trading is very important in economic activities, there are many factors that affect the stock market, and it is also an important indicator of the health of the economic situation (Nguyen et al., 2015). Due to the complexity of the stock market, stock price prediction is one of the most difficult tasks. As stated by Neils Bohr (2011) "Prediction is very difficult, especially about the future". In the 1970s, Fama (1970) proposed the Efficient Market Hypothesis (EMH) has been a major influence in the financial field. According to this theory, the stock market is extremely efficient in reflecting information about individual stocks and the stock market, and it is difficult for investors to predict stock price movements to the extent that it is impossible to make profits beyond those reported by the market average. The Random Walk Theory (RWT) popularised by Malkiel (1973) in "A Random Walk Down Wall Street" further confirmed the EMH by arguing that financial data is irregular and noisy, and that stock market prices move in a random manner, making it impossible to predict the future based on historical data. However, after the 1990s, EMH became less common.

Economists believed that financial crises would not occur if markets were efficient, and many financial analysts and statisticians began to believe that stock prices could be predicted.

2.3 The Stock Historical Data

In this study, there are four predictor variables which are stock historical data. Among them, open price, high price, low price, and volume represent the start value, high value, low value, and volume of the stock for a particular period respectively. The opening price is the value of each share at the opening of trading on the stock exchange, which gives a good indication of the day's movement of the stock, and a significant difference between the previous day's closing price and the day's opening price may indicate news or events affecting the stock. Highs and lows indicate the volatility of the stock during the trading day, and if there is a large gap between them it indicates a high degree of uncertainty about the stock. Volume represents the turnover of the day, which can reflect the liquidity of the stock's market. In summary, these historical stock data have a significant impact on stock prices, and to a certain extent, they can help investors to judge market trends and make trading decisions, and it is important to use them as predictive variables.

2.4 Interest Rate

Many economists and researchers have demonstrated a negative correlation between interest rates and stock prices. Since stock prices represent expected future cash flows, an increase in interest rates increases the opportunity cost of holding money for investors, who tend to hold interest-bearing securities rather than stocks, leading to a decline in stock prices. In addition, changes in interest rates affect investor returns, which in turn affects stock prices (Liu & Shrestha, 2008). From a capital perspective, higher interest rates mean higher borrowing costs for the company, which affects business growth prospects and profits, thus affecting share prices. In conclusion, interest rates are closely related to stock prices and are suitable as predictor variables for this study.

2.5 Inflation Rate

Typically, the cause of inflation is mainly an increase in the money supply or cost-based inflation driven by wages, consumption, and production, etc (Liu & Shrestha, 2008). The relationship between inflation rate and the stock market is elusive, Marshall (1992) showed that a rise in inflationary expectations reduces the expected return on money, which reduces the demand for money and increases the demand for equities, leading to a positive correlation between inflation and stock prices. On the other hand, inflationary expansion weakens the purchasing power of consumers and affects their investment sentiment, adversely affecting stock prices, an argument supported by the study of Geske and Roll (1983). Therefore, this study wants to explore the relationship between inflation rate and stock price and make predictions about future stock prices by using machine learning algorithms.

2.6 Unemployment Rate

So far, there is still uncertainty about the relationship between unemployment rate and stock market, Farsio and Fazel (2013) study rejected the relationship between unemployment rate and stock market, their study mainly focuses on the US, China, and Japan stock markets, and concluded that relying on the unemployment rate to predict the stock price is not accurate. However, on the other hand, Feldmann (2010) concluded that the stock market is closely related to the industrial rate, and that an active stock market reduces the unemployment rate, and that they are negatively correlated. In addition, the industrial rate greatly affects the public's income expectations and investment sentiment. This study will examine the relationship between unemployment rate and stock price and discuss whether it can predict stock price.

With the development of econometric finance, there is a growing interest in analysing and forecasting financial markets. Stock analysis is mainly divided into fundamental and technical analysis. Fundamental analysis is mainly based on analysing the intrinsic value of individual stocks and determining the investment potential in conjunction with

financial conditions and relevant economic data such as financial statements, financial indicators, and industry competitiveness (Wu et al., 2020). Abumostafa & Atiya (1996) argued that there is a price trend in the financial market as well as a certain relationship between the stock prices and economic variables. The study of Singh (2014) and (Chittedi, 2015) showed that money supply, interest rate, inflation rise, exchange rate, gold price and crude oil price have an impact on stock market. Technical analysis is used to predict the future movement of stocks through historical data and technical indicators (Nti et al., 2020). In early applications, some traditional statistical methods were used to predict stock prices such as Simple Moving Average (SMA), Weighted Moving Average (WMA), Exponential Smoothing and Plain Method. Since these statistical methods are linear but stock prices often experience fluctuations and it affects the prediction performance, therefore traditional statistical methods are not so accurate (Grigoryan, 2016). Based on the emerging technologies in finance, financial forecasting applies various machine learning algorithms such as neural networks, decision trees, SVM, plain Bayes, KNN, and random forests, which are capable of handling nonlinear, noisy, and complex data (Kumbure et al., 2022). In addition, some time series models such as Arima, Grach can accurately predict short-term stock trends and returns (Devi et al., 2013). In addition, stock volatility is also affected by several factors such as other indices, political events, and investor sentiment (Nabipour et al., 2020). After the conflict between Russia and Ukraine in 2022, oil supply was affected and investors were worried about the volatility of commodity prices and the S&P 500 index fell by 7% in a few weeks (D. Souza et al., 2022). The stock market is usually sufficiently challenging and changing that forecasting is very difficult and complex, and over the years new techniques and methods have been proposed to try to predict stock trends, giving hope.

2.7 Machine Learning in Stock Prediction

Nowadays, machine learning is widely used in the financial sector. Through machine learning, numerous researchers and financial analysts can process and analyse large amounts of financial data and gain predictive power efficiently and quickly (Bhattacharjee & Bhattacharja, 2019). Machine learning is categorised into supervised and unsupervised learning, some supervised learning algorithms such as Multiple

Regression, SVM, K-Nearest Neighbour (KNN), Random Forest, neural network and Long Short Term Memory (LSTM) are useful for financial market prediction than traditional statistical methods. In addition, according to some scholars, Unsupervised learning algorithms such as K-means have also been shown to be useful in stock market prediction, which can help to identify correlations in stock market data (Shah et al., 2019). Following are the related literatures on Multiple Regression and Random Forest for stock market prediction.

2.7.1 Multiple Regression Model

In statistics, regression analysis is the process of estimating the relationship between dependent and independent variables (Sharma et al, 2017). For stock market analysis, researchers try to use different predictor variables to explore the trend of stock movements and predict future prices.

In the last few decades, it has been widely recognised that government financial policies and macroeconomic events have a significant impact on the stock market. Some researchers wanted to explore their relationship and conducted experiments in different markets. Adam and Tweneboah (2008) also explored the impact of some macroeconomic variables such as FDI, Treasury bill rate, inflation rate and exchange rate on stock prices in Ghana. Their study showed that there is cointegration in the forecasting model and that in the short term period, inflation rate exchange rate has an impact on stocks and in the long run interest rate will be more important. Their research showed that there is a lag in using economic variables to forecast stocks.

Ladan et al. (2014) used Multiple Regression Model to study the relationship between different macroeconomic variables such as inflation rate, exchange rate and interest rate with stock prices. The results of their study showed that exchange rate affects stock prices while interest rate has no significant relationship with stock prices. Also, there is a correlation between independent and dependent variables with R-squared of 54.2%. The limitation of their study is that they only studied three macroeconomic variables

and did not study the impact of more economic variables on stock prices, and it is difficult to prove the linkage of more variables with stock volatility.

Javaid (2010) explored the impact of market variables on price movement in Pakistani stock market using multiple regression model, they collected the historical data of 30 companies in Karachi stock exchange to They collected historical data of 30 companies from Karachi stock exchange and analysed their relationship with KSE 30 index using KIBOR, dividend, earning per share, gross domestic product, and inflation as independent variables. Their study shows that dividend, EPS and KIBOR have positive relationship on stock index while GDP and inflation have negative relationship on stock price. Their model has an accuracy of 62% which proves that some economic variables can accurately predict the stock market.

Gan et al. (2006) used monthly data from 1990 to January 2003 to explore the relationship between seven macroeconomic variables and the New Zealand Stock Index (NZE40). The empirical results proved that NZE40 has a long-run relationship with interest rates, money supply and GDP, and interest rates are negatively correlated with stock prices. In addition, French et al. (1987) revealed the relationship between stock returns and market volatility, and the experimental results show that the expected risk premium of stocks is positively related to the predictive volatility of the market and negatively related to the short-term interest rate.

In addition, the application of machine learning has made technical analysis predictions based on historical stock data increasingly popular. Kamley et al.'s (2013) multiple regression stock prediction model is effective in predicting the monthly movement of stock prices. Their model is based on opening price, closing price and maximum price, where the dependent variable is closing price and independent variables are opening price and maximum price, their model accuracy is 89%, which shows that for stock market prediction, opening price and maximum price can predict the closing price, which to some extent can help investors to get the trend of the stock movement. Bhuriya et al. (2017) use the five predictor variables such as opening price, maximum price,

minimum price, closing price and volume to construct a regression model. The results of their study compared the performance of different regression models, and the result was that the linear regression model was more accurate than the other models with 97%.

Olaniyi et al. (2011) collected daily stock data of three Nigerian banks for 18 months for regression prediction. The special feature of their study is to predict the Current market price by calculating the percentage earnings ratio of the three companies as a dependent variable, which contributes to the banking sector investment in Nigeria.

It is well known that one of the important parameters affecting the volatility of the stock market is the stock trading volume (Rouwenhorst, 2002). It largely reflects the liquidity and price movement of stocks over a certain period. The study of Gharehchopogh et al. (2013) builds a regression model with stock volume as the dependent variable and explores its relationship with average price per share. Their study shows that the variable correlation between Stock Volume and average price per share is 35.8% and the similarity between actual and observed values is found to be 61.35% by testing on 20% validation set data. The study of Seethalakshmi (2018) used six attributes of historical data to build model 1 and then excluded volume and adjusted price from model 2, the result was that model 1 was 5% more accurate than model 2. His study proved that closing price prediction is not affected by adjusted price, but stock volume has a significant impact on the prediction. Therefore, investors and researchers can use stock volume as a reference for stock forecasting.

Some researchers have argued that forecasting monthly returns would provide a more stable and reliable model capable of capturing percentage changes in stock prices over a fixed period. The study by Asghar et al. (2019) et al. created a stock forecasting model where they wanted to predict stock return trends rather than stock prices. Their study analysed the trend of stock returns of three Indian companies from January 2014 to January 2015 by using Monthly simple return as dependent variable and Monthly Volatility, Monthly Volume and Monthly return on investment (ROI) as independent variables. An accuracy of 95%, 89% and 97% is obtained respectively. Thus, monthly

stock volume and monthly ROI are considered as effective predictor variables and their study facilitates the investors to take decisions based on the trend.

2.7.2 Random Forest

Random Forest is a versatile machine learning algorithm that can efficiently perform regression and classification tasks. Compared with other machine learning algorithms, it has excellent performance in long-term trend prediction of financial markets (Tan et al., 2019).

Ballings et al. (2015) used the stock data of 5600 European listed companies to compare the accuracy of different models such as neural network, logistic regression, support vector machine and KNN and random forest for long-term stock price trends. He passed Cross-validation, as an important performance indicator for predicting long-term stock prices, proves that Random Forest has the highest performance.

Milosevic (2016) also applied different machine learning algorithms to stock forecasting. His research proved that random forest has the best performance in predicting long-term price trends compared to algorithms such as svm and naive Bayesian.

Zhang et al. (2018) developed a stock price trend prediction system based on random Forest which can effectively predict the price movement based on the closing price, their study categorised the stock movement into four dimensions: up, down, flat, and unknown, the superior performance of the prediction system gives confidence to the investors.

Lv et al. (2019) collected data from 424 listed companies in the US stock market and 185 listed companies in the Chinese stock market from 2010 to 2017 to compare the performance of different machine learning algorithms with and without transaction

costs. Their study shows that traditional machine learning algorithms have higher performance in the absence of transaction costs.

Tan et al. (2019) developed a stock selection strategy using random forest to analyse the overall performance of A-share companies as well as the industry tendency by analysing the historical data of all the listed companies in the Chinese stock market from 8 February 2013 to 8 August 2017, using the CSI 500 index as the benchmark. At the same time, they got considerable returns, proving the good performance of random forest in the financial market. Finally, they also proved that long-term profits are more related to fundamentals and long-term technical characteristics.

Abraham et al. (2022) used international stock indices to predict stock trends. They used Genetic algorithm and random forest to reveal the link between stock indices and specific stock trends. They predicted the performance of 15 well-known companies and obtained 80 per cent accuracy, and their study concluded that SP500 is the most useful stock index.

Ma et al. (2019) proposed a stock prediction model based on random forest and LSTM neural network to study the volatility of Chinese stock market. They used Random Forest to select data features, and then used 35 technical indicators as features to predict stock movements. The accuracy of RF + LSTM model is 61.18%, which is higher than that of PCA + LSTM and LSTM models. The experimental results show that their prediction model has lower volatility, which is beneficial for investors to get stable returns.

Polamuri et al. (2019) predicted the stocks of all the companies in SP500 index based on the closing price and compared the linear regression, Support Vector Regression, Decision Tree, Random Forest and Extra Tree Regressor, Decision Tree, Random Forest and Extra Tree Regressor. Among them, Decision tree and random forest are the best regression algorithms.

2.8 Time Series in Stock Prediction

The Autoregressive Integrated Moving Average (Arima) model is one of the most widely used models in economics and finance, as well as in stock forecasting (Wadi et al., 2018). However, because of the volatility of the stock market, time series stock market forecasting is considered very challenging (Bao et al., 2017). However, many researchers still try to use the Arima model to predict stock prices and trends, and good results have been obtained, which give great inspiration to this subject.

Devi et al. (2013) selected the 4 most valuable companies in NSE-Nifty Midcap 50 for research. They tested the predictive ability of Arima model in investment decision-making, trying to solve the problems in most stock analysis. Their research verified the model by applying Akaike information criterion (AIC) and Bayesian Information Criterion (BIC) and obtained a very small error rate.

Khan & Alghulaiakh (2020) use Arima model to analyse Netflix five-year historical stock data and make forecasts. Their study considered only the closing price of the stock, as it represents the closing price for the day. At the same time, to make the data smooth, they chose the weekly average data of the stock. Finally, by comparing the different models, Arima (1,1,33) showed the best performance, able to accurately predict the data for the next 100 days. Mondal et al. (2014) used the Arima model for 56 Indian stocks experiment. The proposed model is evaluated using the Akaike Information Criterion (AIC). Experimental results show that the model is effective in improving the prediction accuracy of previous data in different periods.

The research of Jeffrey & Eric (2011) used the Arima intervention analysis method to model the Chinese stock market price index, to evaluate the economic trend of China's stock market price in different periods. This illustrates the predictability of the stock market and the effectiveness of the ARIMA time series model as an analytical and forecasting tool.

According to Pai and Lin (2005), the Arima model can also be used together with other models. For example, in one study, a support vector machine (SVM) was combined with an Arima model to form a hybrid model with linear and nonlinear modelling capabilities to predict stock prices; the hybrid model significantly reduced the overall forecast error and better Predicted the stock price of SBC Communications Inc.

Weng et al. (2022) studied the return on investment of Apple and Tesla, and they concluded that the Arima model is better than Naïve in predicting the change of Tesla's stock price trend. At the same time, their Arima model is $(0,1,1)$, they obtained the ME of the Arima model as 0.000282, which is much smaller than the ME of the Naïve method as 0.836992. The Arima model is often used to solve practical problems in the stock market. By comparing the Arima model with the Naïve method, it can be concluded that the Arima model can predict the overall trend of stock prices and consider the influence of seasonal factors on the prediction results.

Mondal et al. (2014) conducted experiments on 56 Indian stocks using an Arima model. The proposed model is evaluated using the Akaike Information Criterion (AIC). Experimental results show that the model is effective in improving the prediction accuracy of previous data in different periods.

In addition, the Arima model is effective for short-term forecasting. Ariyo et al. (2014) found that Arima model is more robust and efficient than ANN in short-term stock forecasting ability. They constructed the Arima model through the daily closing prices of Nokia and Zenith bank and compared AIC and BIC to obtain the best fitting models of $(2,1,0)$ and $(1,0,1)$ respectively, and obtained a satisfactory result The forecast results, effectively predicting the stock prices of these two companies for several days in the future. Arima models are thus able to compete with emerging forecasting techniques.

Setyo (2017) used the Arima model to predict the Indonesian stock index, which provided a high accuracy rate. The model has a strong short-term forecasting ability

and can compete with the current stock price forecasting technology. Jaya and Amishabaghel (2022) used Python to use machine learning methods for modelling to create a stock forecasting system. They collected a set of time series from May 2018 to May 2019, used the Arima model to predict the future cost of stock, and finally obtained a mape of 98.65%. In other areas of finance, Arima has also demonstrated strong predictive power. Wirawan et al. (2019) collected Bitcoin data from 1 May 2013 to 7 June 2019, and after obtaining an Arima (4,1,1) model with a small mape value through data preprocessing steps such as smoothness testing and differencing, their study successfully predicted the Bitcoin price for the next 7 days. Therefore, these studies largely prove that the Arima model has a strong short-term forecasting ability.

2.9 Research Gap

Quantitative research in finance is becoming popular, and more and more people are exploring the laws of the stock market in the hope of obtaining returns beyond expectations. However, by reviewing the literature, we found that there are many studies on various stock prediction models, but there is a lack of comparative studies and comprehensive analyses between traditional models and machine learning models. This study will focus on analysing the historical stock data and relevant macroeconomic factors of Nasdaq Composite and four well-known companies, namely Tesla, Google, Netflix, and Amazon, with the aim of predicting their stock prices, analysing industry trends and contributing to the financial industry.

2.10 Summary

In this chapter, we review the literature on stock market forecasting as well as time series and machine learning methods. For machine learning algorithm prediction model, both historical stock data and macroeconomic variables can bring predictive power to our model. For time series models, due to the noisy and unstable nature of financial data, data need to be pre-processed and analysed, and it is important for short-term stock forecasting.

Chapter 3. Methodology

This chapter describes the stock forecasting methodology, including the research philosophy, research design, data collection methodology, explanation of predictor variables and introduction of three stock forecasting models. The methodology section aims to provide specific data processing and analysis steps to help the public understand the research methodology of this project. In addition, this research has fully considered the ethical factors that will use the historical stock data from (Yahoo Finance) and the macroeconomic data published by (Statista), which are secondary data and therefore the research is at no risk level.

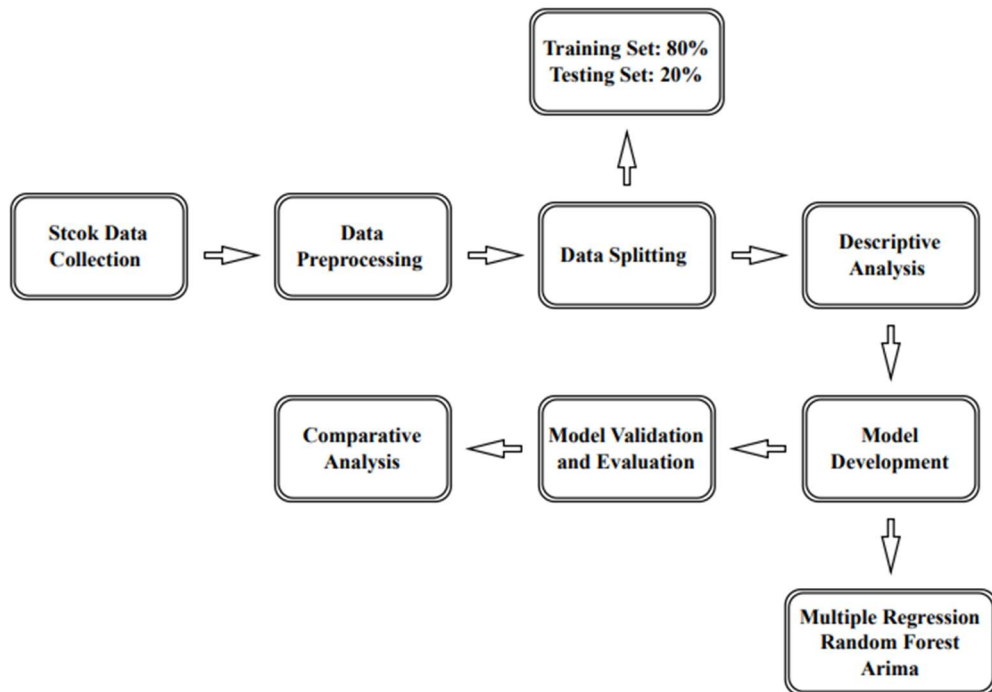


Figure 1. Data Analysis Flow

Figure 1. clearly shows the experimental flow of this study. The first step is to collect monthly historical data from (Yahoo Finance) for Nasdaq Composite and four companies, Tesla, Amazon, Google, and Netflix, containing five variables: opening price, high price, low price, closing price, and volume. In addition, historical data on three macroeconomic variables interest rate, inflation rate and unemployment rate are obtained from (Statista). The second step is data preprocessing, which checks for

outliers and missing values in the dataset and normalises the data before training the model. The third step is to divide the dataset into training set and testing set. The fourth step is to perform descriptive statistics to describe the statistical properties of the variables in the dataset. The fifth step is to build models using time series models and machine learning models and evaluate the performance of three models and finally draw conclusions by comparing the models.

3.1 Research Philosophy

As far as the stock market is concerned, it is a complex dynamic system affected by several factors. The core of this study is to provide a quantitative model for predicting future stock returns. This study is in line with the meaning of positivism, i.e., to build mathematical models such as Multiple Regression, Arima and Random Forest based on objective historical stock data and macroeconomic factors to predict future stock returns and provide inputs to financial institutions and investors in their decision-making.

3.2 Research Tool

This research will use R language to process, analyse and predict large amounts of complex historical stock data and macroeconomic variables. It is an open source programming language with powerful computational and graphical capabilities that is used by data analysts and financial scientists for data analysis, modelling, and visualisation (Kuhn, 2008). R provides a simple, fast, and efficient programming language with many packages to help us analyse the data. For example, 'Forecast' will be used for forecasting, 'tseries' will be used to construct time series, and 'randomForest' will be used to build random forests, which will help us to get accurate results.

3.3 Data Collection

We collected monthly historical stock data for Nasdaq Composite Index and four companies, Tesla, Google, Netflix, and Amazon, from 2020 to 2022 via ticker symbols on a financial website (<https://uk.finance.yahoo.com>).

Table 3.1 Nasdaq Composite Data

Stock	Date	Open	High	Low	Close	Volume
Nasdaq	Jan-2020	9039.46	9451.43	8643.50	9150.94	52694760000
Nasdaq	Feb-2020	9190.72	9838.37	8264.16	8567.37	53886320000
Nasdaq	Mar-2020	8667.14	9070.32	6631.42	7700.10	97320230000
Nasdaq	Apr-2020	7459.50	8957.27	7288.11	8889.55	79171670000
Nasdaq	May-2020	8681.29	9523.64	8537.83	9489.87	81175640000

Table 3.2 TSLA Data

Stock	Date	Open	High	Low	Close	Volume
TSLA	Jan-2020	28.30	43.53	28.11	43.37	6108277500
TSLA	Feb-2020	44.91	64.60	40.77	44.53	7088802000
TSLA	Mar-2020	47.42	53.80	23.37	34.93	6314025000
TSLA	Apr-2020	33.60	57.99	29.76	52.13	5722162500
TSLA	May-2020	50.33	56.22	45.54	55.67	4090554000

Table 3.3 AMZN Data

Stock	Date	Open	High	Low	Close	Volume
AMZN	Jan-2020	93.75	102.79	90.77	100.44	1693966000
AMZN	Feb-2020	100.53	109.30	90.56	94.19	1850202000
AMZN	Mar-2020	95.52	99.82	81.30	94.79	3276182000
AMZN	Apr-2020	96.65	123.75	94.46	123.70	2492196000
AMZN	May-2020	116.84	126.27	122.12	122.12	1651688000

Table 3.4 Netflix Data Sample

Stock	Date	Open	High	Low	Close	Volume
NFLX	Jan-2020	326.10	359.85	321.20	345.09	169874000
NFLX	Feb-2020	347.24	392.95	346.28	369.03	114056000
NFLX	Mar-2020	373.11	393.52	290.25	375.50	201979100
NFLX	Apr-2020	376.05	449.52	357.51	419.85	225111600
NFLX	May-2020	415.10	458.97	397.86	419.73	135055800

Table 3.5 Google Data Sample

Stock	Date	Open	High	Low	Close	Volume
GOOG	Jan-2020	67.08	75.16	67.08	71.71	673832000
GOOG	Feb-2020	73.10	76.61	63.55	66.97	741612000
GOOG	Mar-2020	67.58	70.51	50.68	58.14	142754000
GOOG	Apr-2020	56.10	68.00	53.99	67.43	926702000
GOOG	May-2020	66.43	72.05	64.95	71.45	63732600

From Table 3.1 - 3.5, the dataset has five different attributes such as opening price, high price, low price, closing price and volume.

Table 3.6 Macroeconomic Data Sample

Date	Inflation Rate (%)	Interest Rate (%)	Unemployment Rate (%)
Jan-2020	2.50	1.55	3.5
Feb-2020	2.30	1.58	3.5
Mar-2020	1.50	0.65	4.4
Apr-2020	0.40	0.05	14.7
May-2020	0.20	0.05	13.2

From Figure 3.6, we can understand the data of three major macroeconomic variables such as monthly interest rate, monthly inflation rate, and monthly unemployment rate from 2020 to 2022 for the United States.

3.4 Data preprocessing

Data preprocessing is an important part of data analysis. Before conducting the experiment, we need to convert the raw data into input data to improve the efficiency and accuracy of data analysis. Firstly, as financial data are usually very complex and incomplete, we need to use Excel to deal with the missing values and outliers in the original dataset. Secondly, the data needs to be pair-normalised to reduce the scale sensitivity of the model and increase the speed of convergence. Finally, the data is split

into a training set and a 20% test set in chronological order, and the test set is used to test the performance of the model to avoid overfitting.

3.5 Predictive Model Development

After converting the raw data into input data, the data was divided into 80% training set and 20% test set. The training set is used to build the model and the test set is used to test the performance of the model.

3.5.1 Linear Regression

Linear regression belongs to the supervised learning models in machine learning that can show the relationship between dependent and independent variables and predict the future (Panwar et al., 2021). Its formula is (2), where Y is the dependent variable, β_0 is the intercept, β_1 is the correlation coefficient of the independent variable X , and ϵ represents the residuals of the regression model. The model assumes that the residuals exhibit a positive too distribution and that there is no high correlation between the independent variables. Specifically, regression analysis helps to understand how the dependent variable changes when the independent variable changes and use that variable to make numerical predictions (Seber & Lee, 2012).

$$Y = \beta_0 + \beta_1 X + \epsilon \quad (2)$$

This study used multiple linear regression model to analyse the relationship between stock close price and seven variables such as open price, high price, low price, volume, interest rate, inflation rate and unemployment rate, and aims to find the linear combination with the best predictive results. The formula is as follow:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \beta_6 X_6 + \beta_7 X_7 + \epsilon \quad (3)$$

Where,

Y = Close Price

X_1 = Open Price

X_2 = High Price

X_3 = Low Price

X_4 = Volume

X_5 = Interest Rate

X_6 = Inflation Rate

X_7 = Unemployment Rate

3.5.2 Random Forest

Random Forest is another machine learning supervised model in this study, which is an integrated approach consisting of multiple decision trees that improves prediction accuracy and can be used for regression and classification tasks. As the complexity of financial data tends to cause overfitting in traditional decision tree models, Random Forest is relatively robust to outliers and noise, and it overcomes this problem by training multiple decision trees on different subspaces of the feature space (Khaidem et al., 2016). Belgiu and Dragut (2016) in their literature review showed that RF classifier can handle high dimensional data and multiple covariates. In addition, Random Forest algorithm provides the best segmentation features based on random feature selection and constructs a decision tree, which usually grows to maximum depth without pruning. The output is the set of all decision trees and for regression problems, the predictions of each tree need to be averaged to get the result. This study aims to obtain excellent stock price prediction model by modelling 8 variables with the powerful performance of RF.

3.5.3 Arima Model

The main objective of time series analysis is to understand the underlying structure and patterns of the data and develop models to predict future values. Arima Model assumes that the future of a variable is just a linear combination of past tense and past error and provides good short-term forecasting ability for investors in the financial sector (Khandelwal & Mohanty, 2021). The model is divided into three components:

Autoregressive (AR): Indicates the relationship between current and historical values, using the variable's historical data to forecast itself.

Integration (I): Indicates the order of difference required to make the series stationary.

Moving Average (MA): Indicates the cumulative value of the error term in the autoregressive model.

The formula of Arima model:

$$Y_t = c + \phi_1 * y'(t-1) + \dots + \phi_p * y'(t-p) + \theta_1 * \epsilon(t-1) + \dots + \theta_q * \epsilon(t-q) + \epsilon_t \quad (4)$$

Where,

Y_t represents the actual value, C is a constant, ϵ_t represents the random error at t , ϕ_i and θ_i are the coefficients, p is the lag order of the autoregressive part, and q is the lag order of the moving average part.

In this study, we will model and analyse and forecast the closing prices for Nasdaq Composite and four companies which is Tesla, Google, Netflix, and Amazon, for 36 months from 2020 to 2022. The experimental procedure is divided into three parts: first,

we perform smoothness test and white noise test on the dataset to confirm whether the difference order and the series fluctuate randomly. Second, the Acf and Pacf of the dataset are analysed to obtain the best p-values and q-values. Finally, the Arima model is built, and model testing is performed to obtain prediction results.

3.6 Description of Model Evaluation Metric

After modelling the training sets, we need several key metrics to evaluate their performance, here is a description of the performance evaluation metrics.

Mean Absolute Error (MAE):

MAE represents the absolute error produced by the prediction model in the predicted data, which is achieved by calculating the absolute difference between the predicted and actual values and calculating the mean. When MAE is expressed as 0, the model is perfect, and its higher value indicates a higher error between the predicted and actual values.

Root Mean Square Error (RMSE):

RMSE represents the square root of the mean squared difference between the predicted and actual values. Similar to MAE, when its value is 0, the predictive model is perfect. The smaller its value, the better the model performance.

R-squared (Coefficient of Determination):

R-squared represents the proportion of the variance of the dependent variable that can be predicted from the independent variable. When R-squared is 1, the predictive model is perfect, and when it has a value of 0, the model is not linear. Therefore, a higher R-squared indicates a better performance of the predictive model, but there is a need to be concerned about whether the model is overfitted.

Cross-validation:

This study will also test the performance of the model using cross-validation. Since the stock data is a financial time series and has a fixed time structure, the traditional cross-validation is not available. In the data processing part, the dataset needs to be partitioned by time and the model needs to be trained several times to obtain the model performance. It is beneficial to reduce overfitting as well as to obtain more reliable performance estimates.

3.7 Ethical Issue

The research has received ethical approval from The University of Sheffield. At the same time, this research will use second-hand data as the data source, and all data will be downloaded from public websites, which does not involve the use of personal data and information. Finally, research dataset will be stored entirely on personal or school computers.

Charter 4. Findings and Discussion

This charter focuses on presenting the results of modelling the prediction models, comparing the prediction results of different models on the dataset and comparing the prediction accuracies of the different models with the aim of visually presenting the experimental results of this study and providing the readers with a methodology for stock price prediction.

4.1 Predictive Modelling Result

This study will examine the closing prices of Nasdaq Composite and four companies with a total of 36 observations per dataset. The Nasdaq Composite, as a stock market index that comprehensively reflects stock market trends, is an important indicator for the technology sector and is important for the financial sector. Figure 2. shows its performance from January 2020 to December 2022 with a general upward and downward trend. Figure 3. shows the stock movements of the four giants of the technology sector, with Tesla and Netflix showing similar fluctuations to the Nasdaq, while Amazon and Google remained flat.

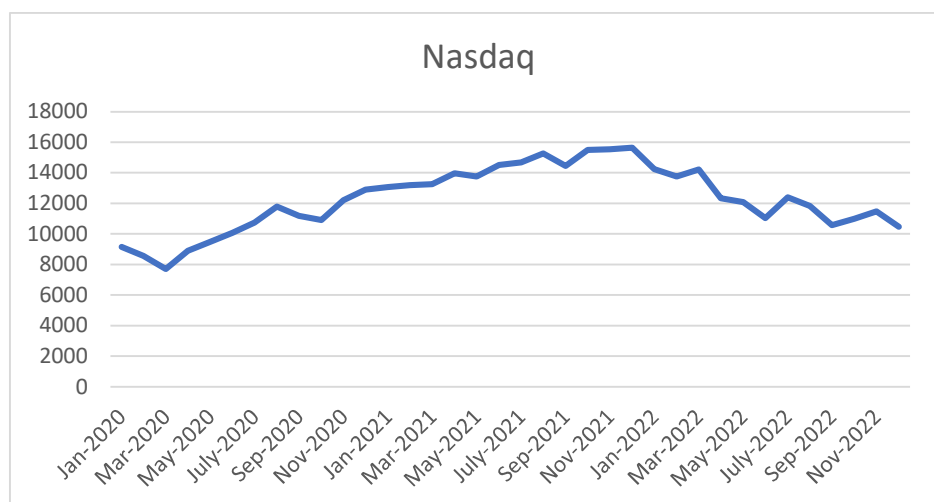


Figure 2. Nasdaq Composite Close Price

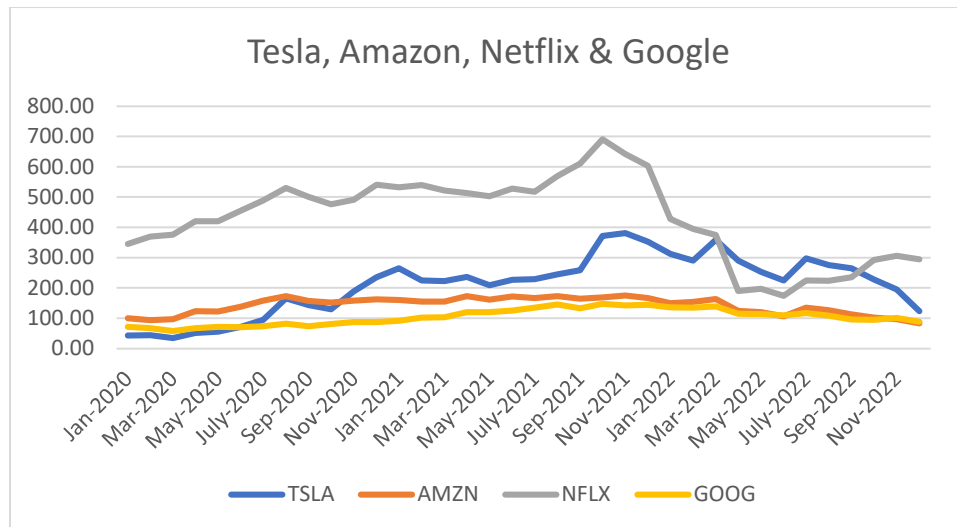


Figure 3. Close Price for Tesla, Amazon, Netflix, and Google

4.1.1 Linear Regression

Model 1:

	Estimate	Standard Error	t value	Pr(> t)	
(Intercept)	0.124	0.088	1.405	0.1746	
Open	-0.577	0.248	-2.323	0.0303	*
High	0.560	0.304	1.842	0.0796	.
Low	0.864	0.188	4.603	0.0002	***
Volume	-0.008	0.047	-0.170	0.8666	
IR	-0.387	0.244	-1.588	0.1272	
Inflation.rate	0.108	0.061	1.775	0.0904	.
UNRATE	-0.049	0.100	-0.489	0.6301	

Signif. codes: 0 '***' < 0.001 < '**' < 0.01 < '*' < 0.05

Residual standard error: 0.04677 on 21 degrees of freedom

Multiple R-squared: 0.9803, Adjusted R-squared: 0.9737

F-statistic: 149.1 on 21 and 7 DF, p-value: 0.0000

Figure 4. Summary of Nasdaq

VIF Test:

Table 4.1 VIF Test for Nasdaq

Open	High	Low	Volume	Interest Rate	Inflation Rate	Unemployment Rate
67.76	116.10	31.66	1.65	7.54	4.83	8.45

According to Figure 4., Model 1 shows the experimental results of the regression model for the Nasdaq Composite Index, and the correlation between each predictor variable and the response variable can be examined through the t-test. The P-value of opening price is 0.0303, which is weakly correlated with multicollinearity; the P-value of low price is 0.0002, which is strongly correlated with closing price. Due to the characteristics of financial data, the stock historical price usually has a fixed trend and closing dependence, and is affected by macroeconomic factors, financial conditions and other fundamental factors, there may be multicollinearity among the stock historical data. According to Table 4.1, the variance expansion factor values of the opening, high and low prices are all greater than 10, so it can be judged that there is multicollinearity among them. multicollinearity.

Model 2:

	Estimate	Standard Error	t value	Pr(> t)
(Intercept)	0.739	0.139	5.332	0.0000 ***
Volume	-0.003	0.135	-0.022	0.9830
IR	-2.067	0.381	-5.431	0.0000 ***
Inflation.rate	0.257	0.128	2.010	0.0558 .
UNRATE	-0.582	0.179	-3.256	0.0034 **

Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05

Residual standard error: 0.1398 on 24 degrees of freedom

Multiple R-squared: 0.7986, Adjusted R-squared: 0.765

F-statistic: 23.79 on 24 and 4 DF, p-value: 0.0000

Figure 5. Summary of Nasdaq

According to Figure 5., Model 2 was developed to observe the significance and correlation of volume, interest rate, inflation rate and unemployment rate. The result shows that volume is not significant, while interest rate, inflation rate and unemployment rate show significance. Also, Volume, Interest rate and unemployment rate are negatively correlated with close price while inflation rate is positively correlated with close price, demonstrating the same correlation with the full model. In conclusion, the presence of multicollinearity in the variables may make the estimated coefficients unstable, but it does not have much negative impact on the performance of the prediction model, and the full regression model can be effective in predicting the stock price and comparing it with other models.

From the above analyses, the following linear equation was developed for five datasets.

Nasdaq:

$$\begin{aligned} \text{Predicted Close Price} = \\ 0.124 - 0.577*(\text{Open}) + 0.56*(\text{High}) + 0.864*(\text{Low}) - 0.008*(\text{Volume}) - \\ 0.387*(\text{Interest Rate}) + 0.108*(\text{Inflation Rate}) - 0.049*(\text{UNRATE}) \end{aligned} \quad (5)$$

Tesla:

$$\begin{aligned} \text{Predicted Close Price for} = \\ 0.044 - 0.647*(\text{Open}) + 0.837*(\text{High}) + 0.646*(\text{Low}) + 0.076*(\text{Volume}) - \\ 0.452*(\text{Interest Rate}) + 0.125*(\text{Inflation. Rate}) - 0.097*(\text{UNRATE}) \end{aligned} \quad (5)$$

Amazon:

$$\text{Predicted Close Price} =$$

$$0.174 - 0.457*(\text{Open}) + 0.43*(\text{High}) + 0.785*(\text{Low}) + 0.035*(\text{Volume}) - 0.443*(\text{Interest Rate}) + 0.081*(\text{Inflation. Rate}) - 0.005*(\text{UNRATE}) \quad (5)$$

Netflix:

Predicted Close Price =

$$0.183 - 0.564*(\text{Open}) + 0.945*(\text{High}) + 0.339*(\text{Low}) - 0.293*(\text{Volume}) - 0.037*(\text{Interest Rate}) + 0.024*(\text{Inflation. Rate}) - 0.001*(\text{UNRATE}) \quad (5)$$

Google:

Predicted Close Price =

$$-0.015 - 0.759*(\text{Open}) + 0.589*(\text{High}) + 1.08*(\text{Low}) + 0.073*(\text{Volume}) - 0.069*(\text{Interest Rate}) + 0.112*(\text{Inflation. Rate}) - 0.023*(\text{UNRATE}) \quad (5)$$

4.1.2 Random Forest

Table 4.2 Random Forest of Nasdaq

Number of trees:	500
No. of variables tried at each split:	7
Mean of squared residuals:	0.008332672
% Var explained:	89.62

Table 4.2 shows the Random Forest modelling process based on the Nasdaq dataset. Firstly, the training model will create 500 decision trees to increase its stability. Then, through cross-validation, we were able to determine that the model achieves the best predictive performance when the value of mtry is 7, which means that when each decision tree is split, 7 variables are considered as features that can reduce bias. Finally, the MSE of the proposed model is 0.008332672 and Var explained is 89.62%, which means that the model has very little difference between the observed and predicted values and has a better predictive ability.

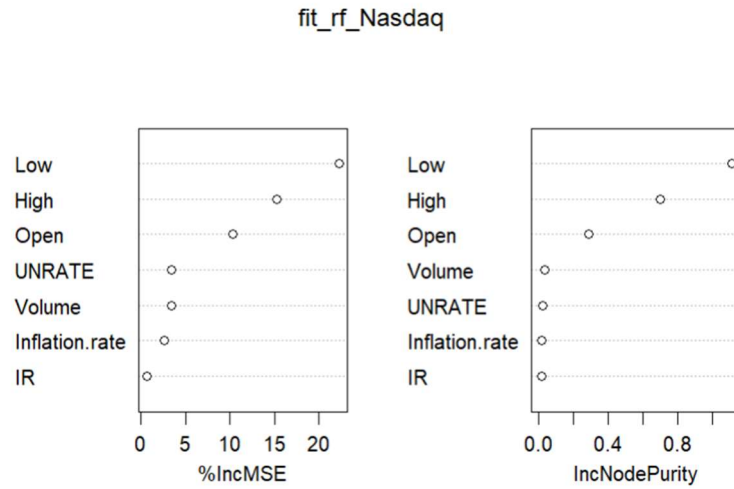


Figure 6. Random Forest of Nasdaq

Figure 6. shows the results of the data analysis obtained from modelling using Random Forest based on the Nasdaq dataset. Both IncMSE and IncNodePurity are key indicators of the importance of a variable, with higher observed values indicating that the variable is more important in the prediction. Where IncMSE indicates the error variation of the prediction model and IncNodePurity indicates the purity of the nodes used to create the decision tree. The experimental results show that in predicting the Nasdaq closing price, low price is the most important feature in the prediction model, followed by high price, while trading volume has little effect. Among the macroeconomic variables, the unemployment rate is the key feature in predicting stock Nasdaq close price, followed by inflation rate, while the inflation rate has little effect. In addition, historical stock data contribute more to stock price forecasting than macroeconomic variables.

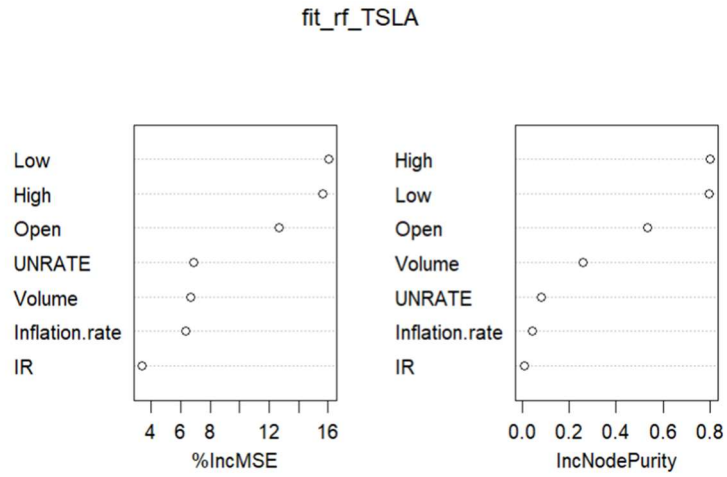


Figure 7. Random Forest of Tesla

Figure 7. shows the importance of Tesla's features in random forest modelling. In the stock historical data, High price and Low price are important for model prediction. Among the macroeconomic variables, Inflation rate has the greatest impact on Tesla stock price prediction and Interest rate has the least impact.

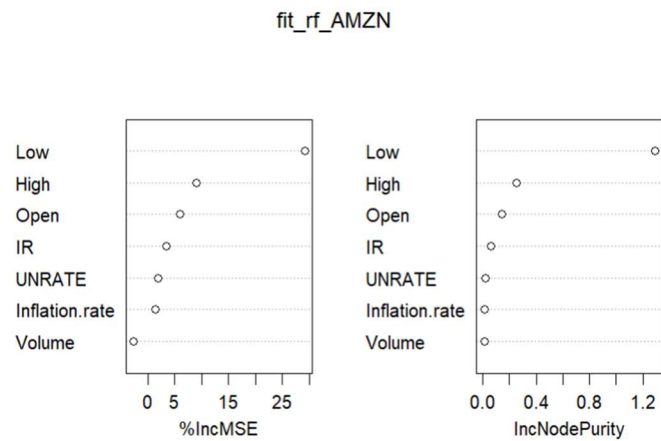


Figure 8. Random Forest of Amazon

Figure 8. shows the importance of Amazon's features in random forest modelling. In the stock historical data, High price and Low price are important for model prediction. Among the macroeconomic variables, the interest rate has the greatest impact on Amazon's stock price prediction and the inflation rate has the least impact.

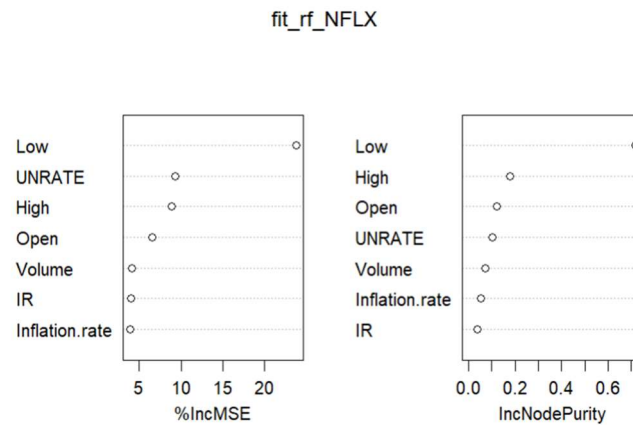


Figure 9. Random Forest of Netflix

Figure 9. shows the importance of Netflix features in Random Forest modelling. Unlike the dataset between, Low Price and the unemployment rate have a greater impact on the predictive model, and the Inflation rate and the Interest Rate have a smaller impact.

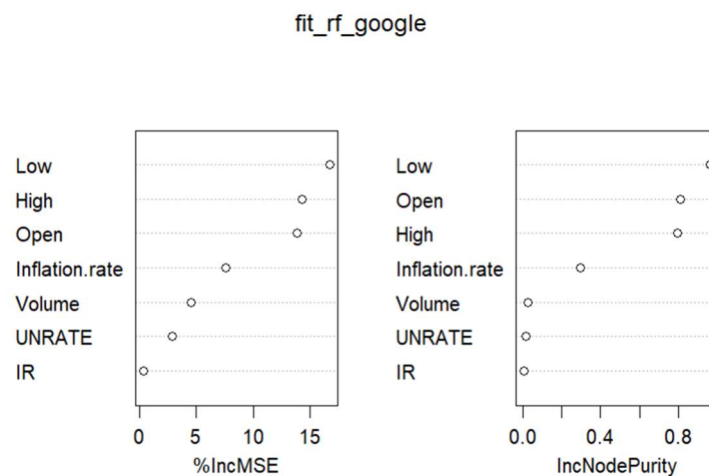


Figure 10. Random Forest of Google

Figure 10. shows the importance of Google's features in Random Forest modelling. Historical stock data has a significant impact, while among macroeconomic variables, Inflation rate demonstrates higher Increased in Mean squared error and purity.

4.1.3 Arima

R language was able to utilise the (Auto.arima) function and implement the best Arima model selection based on AIC and BIC criteria. It automatically selects the parameters of the Arima model and gives a reasonable prediction model.

```
> fit = auto.arima(fit_Nasdaq_ts, stepwise=TRUE, trace=TRUE,
ARIMA(2,1,2)          with drift      : Inf
ARIMA(0,1,0)          with drift      : 455.7334
ARIMA(1,1,0)          with drift      : 458.2532
ARIMA(0,1,1)          with drift      : 458.2532
ARIMA(0,1,0)          with drift      : 453.9272
ARIMA(1,1,1)          with drift      : 460.8097
Best model: ARIMA(0,1,0)

> fit = auto.arima(fit_tesla_ts, stepwise=TRUE, trace=TRUE,
ARIMA(2,1,2)          with drift      : Inf
ARIMA(0,1,0)          with drift      : 288.6808
ARIMA(1,1,0)          with drift      : 291.1822
ARIMA(0,1,1)          with drift      : 291.167
ARIMA(0,1,0)          with drift      : 287.3805
ARIMA(1,1,1)          with drift      : 292.4998
Best model: ARIMA(0,1,0)

> fit = auto.arima(fit_amazon_ts, stepwise=TRUE, trace=TRUE)
ARIMA(2,1,2)          with drift      : Inf
ARIMA(0,1,0)          with drift      : 227.9371
ARIMA(1,1,0)          with drift      : 230.3911
ARIMA(0,1,1)          with drift      : 230.3773
ARIMA(0,1,0)          with drift      : 225.6924
ARIMA(1,1,1)          with drift      : 233.0939
Best model: ARIMA(0,1,0)

> fit = auto.arima(fit_google_ts, stepwise=TRUE, trace=TRUE)
ARIMA(2,1,2)          with drift      : Inf
ARIMA(0,1,0)          with drift      : 205.4551
ARIMA(1,1,0)          with drift      : 206.9087
ARIMA(0,1,1)          with drift      : 207.1255
ARIMA(0,1,0)          with drift      : 203.9512
ARIMA(1,1,1)          with drift      : 209.6055
Best model: ARIMA(0,1,0)
```

Figure 11. The Arima result of Nasdaq, Tesla, Amazon, Google

According to Figure 11., the best model parameters for Nasdaq Composite, Tesla, Amazon, and Google are all (0,1,0), showing that their closing prices are random wandering sequences, implying that the new observation is a random error of the previous one, and that there is no clear trend or seasonal model.

```
> fit = auto.arima(fit_Netflix_ts, stepwise=TRUE, trace=TRUE)

ARIMA(2,0,2)          with non-zero mean : 327.7829
ARIMA(0,0,0)          with non-zero mean : 360.4394
ARIMA(1,0,0)          with non-zero mean : 324.961
ARIMA(0,0,1)          with non-zero mean : 341.2796
ARIMA(0,0,0)          with zero mean      : 443.4407
ARIMA(2,0,0)          with non-zero mean : 325.3015
ARIMA(1,0,1)          with non-zero mean : 326.1025
ARIMA(2,0,1)          with non-zero mean : Inf
ARIMA(1,0,0)          with zero mean      : 325.1465

Best model: ARIMA(1,0,0)          with non-zero mean
```

Figure 12. The Arima result of Netflix

According to Figure 12., which is Arima model result of Netflix. Its parameter is Arima (1,0,0), and its AIC value is 325.15, which means that the previous observation is significant for the prediction value, and it can predict based on the previous observation.

4.2 Prediction

After modelling the dataset, an important step is to apply the predictive model on a test set to confirm the accuracy of the predictive model. Below are the results of the Nasdaq Composite, Tesla, Amazon, Netflix, and Google stock price predictions based on Linear Regression, Random Forest and Arima.

Nasdaq Composite:

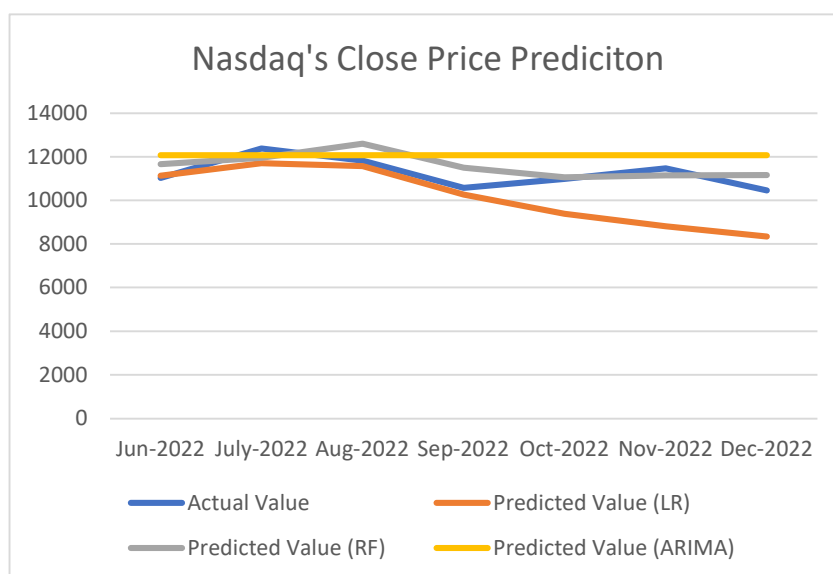


Figure 13. The Prediction Result of Nasdaq

Figure 13. shows the predicted prices of the three models for the closing prices of Nasdaq from June 2022 to December 2022. For the first month ahead, Linear Regression has the strongest short-term predictive performance with a residual of 96.2, which achieves a strong prediction relative to the Random Forest at 634.9 and the Arima Model at 1052.65. The Nasdaq Composite under the Arima Model is a random walk series and hence it has only one month's prediction result which is the least accurate among the three models. Random Forest effectively predicts the volatility of the stock

price and has strong predictive power in long term forecasting relative to Linear Regression.

Tesla:

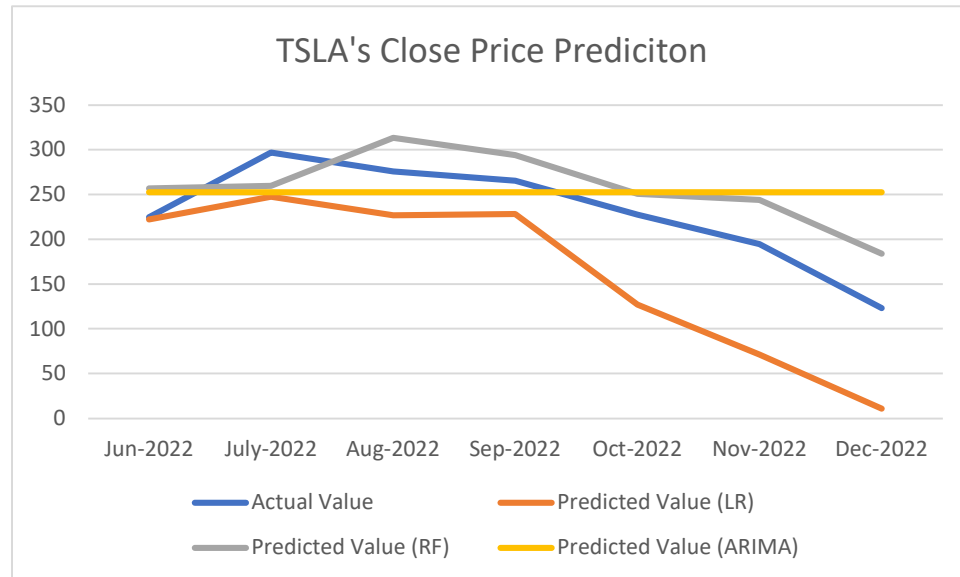


Figure 14. The Prediction Result of Tesla

According to Figure 14., in Tesla's stock price prediction, Linear Regression obtained the smallest residuals in the first month, while Random Forest was less accurate. The further the period goes on, the better the Random Forest's predictive ability becomes, basically keeping in line with the stock volatility trend.

Amazon:

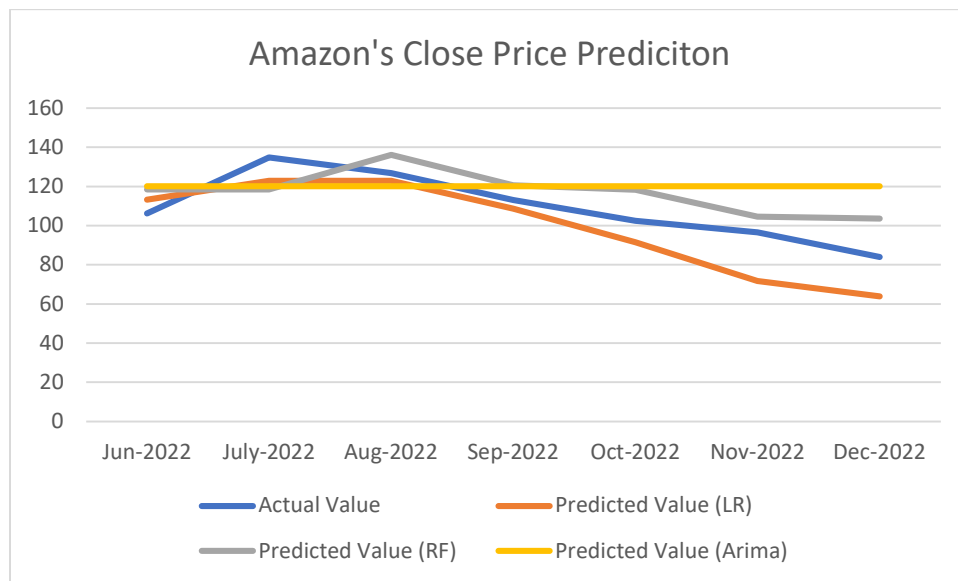


Figure 15. The Prediction Result of Amazon

According to Figure 15., the three prediction models show high prediction accuracy in Amazon's June 2022 closing price prediction, with Linear Regression having the highest accuracy. Subsequently, as in the above analysis, Random Forest can predict stock price fluctuations first, but with a lag.

Netflix:

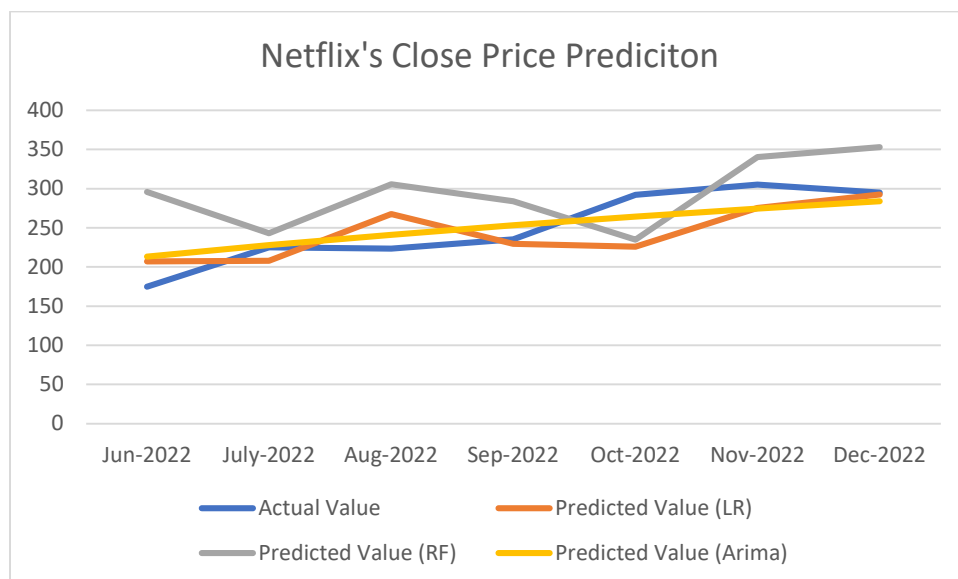


Figure 16. The Prediction Result of Netflix

According to Figure 16., Linear Regression continues to be the strongest performer in the first month of predictions. However, the Arima Model is the strongest predictor of Netflix for seven months, with a value of 20.828 for its MAE, compared to Linear Regression's 28.2 and Random Forest's 59.91. The Arima model can predict the volatility of Netflix's stock price very well.

Google:

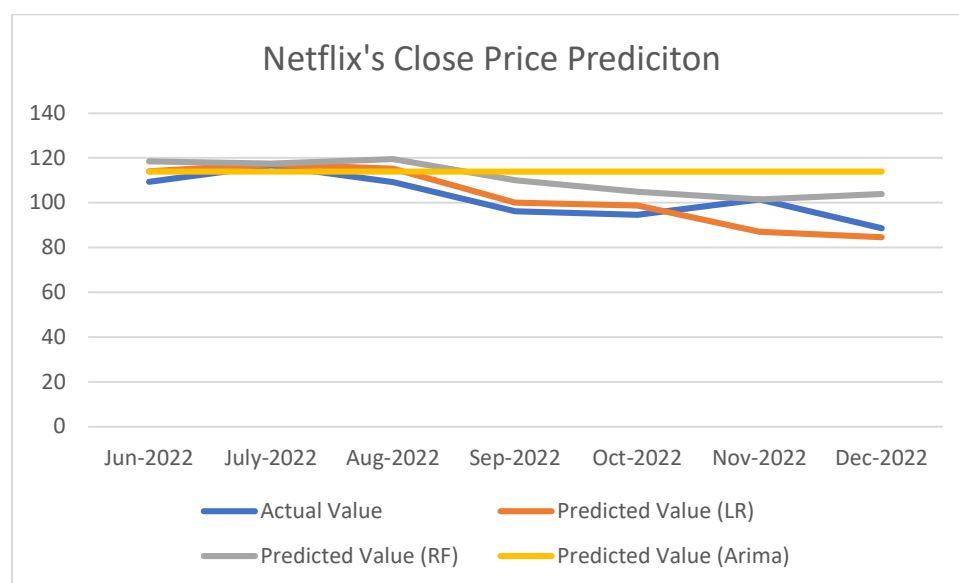


Figure 17. The Prediction Result of Google

According to Figure 17., Linear Regression, Random Forest and Arima have high accuracy in predicting Google's stock price in the next three months. In addition, Linear Regression can predict the volatility of the stock price and give accurate investment decision to the investors.

4.3 Evaluation on forecasting Results

Prediction accuracy demonstrates the validity of a model's ability to predict future stock market ups and downs. After training the model in R, the MAE, RMSE and R-squared are calculated using the prediction results and the test set, all of which can be used to

test the performance of the model. R-squared is the core indicator, it shows the extent to which the predictor variables can explain changes in stock returns (Nagar and Hahsler, 2012).

Stock	Model	MAE	RMSE	R-Squared
Nasdaq	Linear Regression	1102.49	1451.87	0.90
	Random Forest	552.95	610.95	0.98
	Arima	922.06	1050.22	0.95

Figure 18. Model Evaluation of Nasdaq

Nasdaq Composite is an index of more than 3,000 stocks, mainly made up of technology stocks, so predicting its fluctuations can help investors gain insight into the industry's outlook, and accurate predictions can be used in trading systems to make buy or sell decisions based on the market's movements.

Figure 18. shows the experimental results of this study on Nasdaq Composite, where Random Forest achieves significant performance accuracy in Nasdaq stock price prediction. By comparing it with Linear regression and Arima, it obtained 98% R-squared and smaller MAE and RMSE. This means that Random Forest's prediction residuals are smaller compared to other models in the test set. Due to the complexity of financial markets, forecasting is very difficult, and no model is perfect. Based on the characteristics of predictive models, linear regression assumes that there is a linear relationship between the independent variables and the variables, and multicollinearity is an important factor that affects the predictive ability when the predictor variables contain historical stock data. In addition, the Arima model is good at analysing and forecasting time series data with smoothness, using past values and error values to predict the future. Its predictive performance is poor if the data set is not sufficiently volatile or is affected by external factors such as market sentiment, news, etc. However, Random Forest has excellent feature filtering performance and can handle non-linear relationships between variables. Therefore, in this study, based on the historical stock data of open price, high price, low price, and volume and the macroeconomic variables interest rate, inflation rate, and unemployment as predictor variables, Random Forest

has the best prediction performance in predicting the long-term trend based on multiple variables. Random Forest has the best prediction performance based on multiple variables and predicting long-term trends.

			MAE	RMSE
Nasdaq	1 Month	Linear Regression	96.20	96.20
		Random Forest	634.90	634.90
		Arima	1052.65	1052.65
	3 Month	Linear Regression	594.58	421.64
		Random Forest	760.71	1118.82
		Arima	802.65	1052.65
	7 Month	Linear Regression	1102.49	1451.87
		Random Forest	552.95	610.95
		Arima	922.06	1052.22

Figure 19. Model Evaluation of Nasdaq

According to Figure 19., for the Nasdaq dataset, Linear Regression obtains the best forecasting performance in one-month and three-month forecasting, and its MAE and RMSE are smaller than those of Arima and Random Forest, which proves that it has excellent short-term prediction capability. In addition, for the seven-month prediction, Random Forest has lower MAE and RMSE, and it is able to capture trends and patterns in the data and is less prone to overfitting, which gives it a better performance in the long-term prediction.

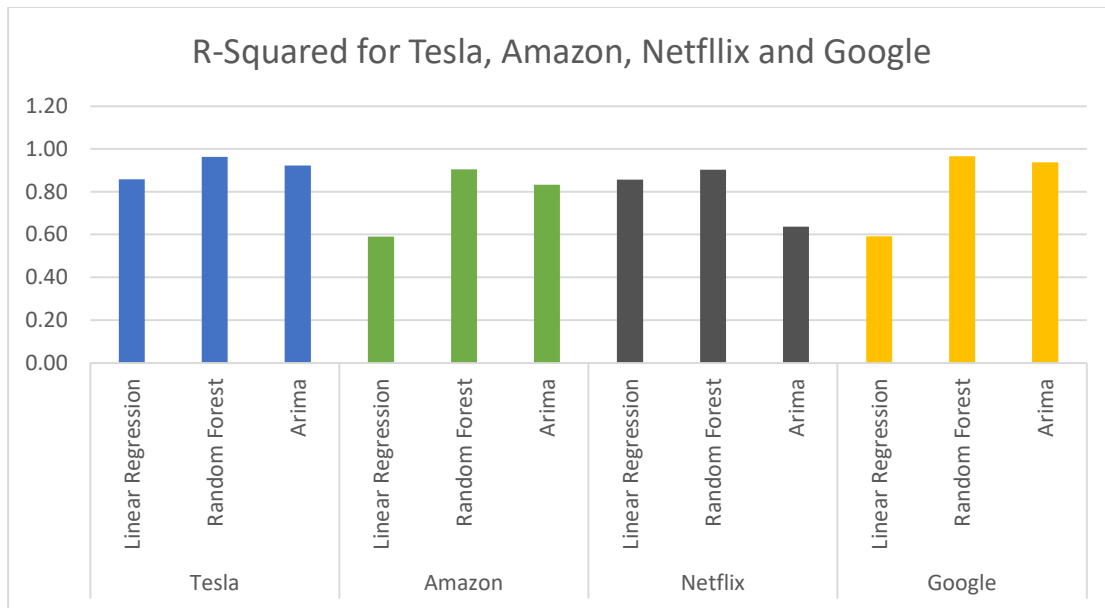


Figure 20. Model Evaluation of Tesla, Amazon, Netflix and Google

Figure 20. shows that for all four Nasdaq-listed companies, the Random Forest-based predictions produce the best predictive power. For prediction, the higher the R-squared which with the better predictive model. Therefore, it can be demonstrated that investors are able to capture long-term trends using random forest, while stock historical data such as Open Price has little impact on the company's stock price prediction.

4.4 Discussion

The following will discuss the findings of this study and compare them with those of previous studies and provide explanations.

4.4.1 Research Question 1

How is the seven predictor variables affect the Close price based on the dataset?

The financial market is a volatile market, and its fluctuations are affected by a variety of factors, scholars in the field of finance are constantly trying to explore the variables

affecting the stock market, with the aim of exploring the laws of the stock market. In the empirical analysis section, this study analyses the relationship between seven variables, i.e., Open Price, High Price, Low Price, Volume, Interest Rate, Inflation Rate and Unemployment Rate, and the closing price through linear regression and Random Forest. closing price. Broad market indices usually reflect industry trends and market sentiment, or even the overall economy, while the share prices of individual company stocks reflect the market expectations of the firms.

Similar to the studies of (Kamley et al., 2013; Bhuriya et al., 2017), Open Price is negatively correlated with Close Price, while high price and low price are positively correlated. This is because these variables provide the trend of price fluctuations on the day and are important investment indicators, thus affecting investor sentiment. The Arbitrage Pricing Theory (APT) proposed by Ross (1976) explains the explanatory factors of asset returns, and changes in macroeconomic variables can be viewed as changes in potential systematic risk factors that affect future returns. For the analysis of the three macroeconomic variables in this study, similar to the studies of (Gan et al., 2006; Humple & Macmilan, 2009), interest rate and unemployment rate are negatively correlated with stock prices, while the inflation rate is positively correlated with stock prices. In terms of the US stock market, Jareno and Negrut (2015) empirical study of the Dow Jones Industrials and the S&P 500, the stock market has a strong positive correlation with the CPI index, GDP, and IPI, and a negative correlation with the interest rate and the unemployment rate. In addition, Humple and Macmilan's study concludes that interest rate affects the stock prices in the US and Japan and is negatively correlated. Therefore, the study based on the Nasdaq Composite and four U.S. companies can explain the relationship between macroeconomics and the stock market. In terms of studies on other economies, Nasseh and Strauss (2000) explored the relationship between macroeconomic variables and the stock markets of six European economies, revealing that there is a significant long-run relationship between long-run interest rates, consumer prices and manufacturing production and stock prices.

4.4.2 Research Question 2

Which stock price prediction model has the best performance?

In this study, five datasets, Nasdaq Composite, Tesla, Amazon, Netflix, and Google, are modelled based on Linear Regression, Arima Model and Random Forest and the models are evaluated by metrics such as MAE, RMSE, and R-squared with the aim of discover the model with the best prediction performance. The empirical results demonstrate that Random Forest has the highest prediction accuracy for the test data sets, 98% for Nasdaq, 96% for Tesla, 91% for Amazon, 90% for Netflix, and 97% for Google, all of which are preferred to Linear Regression and Arima model. Regression and Arima model. Similar to the findings of the earlier studies (Tan et al., 2019; Ballings et al., 2015; Milosevic, 2016), this is due to the fact that Random Forest is capable of filtering and processing complex models when dealing with a large number of features and effectively captures nonlinear relationships, reducing the risk of model overfitting. However, compared to Linear Regression and Arima, Random Forest has a higher computational complexity due to its splitting by decision tree features and slower computational speed when dealing with complex predictive models.

In addition, the prediction performance of Linear Regression, Arima and Random Forest is compared over time. The results of the study show that Linear Regression performs best in short-term prediction. By comparing with the previous studies (Wazarkar et al., 2022; Bhuriya et al., 2017; Kamley et al. ,2013), it is found that they tend to model and make predictions from historical data such as Open Price, High Price and Low price, ignoring the significant external factors on stock impact on stocks. In contrast, our approach combines stock data and external factors as independent variables, which improves the predictive performance of the model.

However, the Arima Model does not demonstrate good predictive results in this study, and the modelling results for four datasets such as Nasdaq present a random wandering sequence, a finding that differs from the findings of (Ariyo et al. ,2014; Setyo, 2017). By comparing the methodological differences, we find that the different data sample sizes may be an important reason for the difference in forecasting performance, too few

stock period data do not allow Arima to capture the autocorrelation of stock prices. In addition, financial data is dynamic, complex, and non-linear, and it can be affected by many external factors, such as government decisions, special events or market sentiment. The Arima Model may be better at predicting time series data that are autocorrelated, free of external factors and smooth.

Charter 5. Conclusion

5.1 Summary

Predicting the future is often fascinating. In recent years, machine learning techniques have proved to be effective in forecasting in financial markets. Forecasting plays an important role in the stock market due to its dynamic nature, however it is a very difficult and challenging process. The objective of this thesis is to study the effectiveness of traditional time series model Arima and machine learning models linear regression, Random Forest in predicting the closing price of Nasdaq Composite, Tesla, Amazon, Netflix, and Google stocks and to understand the relationship between historical stock data and macroeconomic variables and the stock prices. It is hoped to provide clear explanations to relevant stakeholders in the financial field and to advance the role of relevant technology in financial decision-making.

Firstly, by comparing the stock price prediction performance of Linear Regression, Random Forest and Arima, Random Forest performs the best in predicting the long-term stock price trend, and its R-squared is significantly higher than that of the other two models in all five test sets. In addition, by comparing the prediction accuracy of the three prediction models at one month, three months and seven months, it is found that Linear Regression is able to obtain lower error in short-term prediction. By comparing the predictive performance of different datasets, all three predictive models show good predictive validity in Nasdaq, which may be due to the fact that the Nasdaq Composite contains the stocks of many companies, which represent the industry trend and the general economic performance, and the stock price fluctuation is more stable and sensitive to the changes of macroeconomic variables. This may be due to the fact that the Nasdaq Composite contains stocks of many companies that represent industry trends and general economic performance, which are more stable and sensitive to changes in macroeconomic variables.

Through the correlation analysis of the variables in the five datasets, it is found that Open Price is negatively correlated with Close Price, while High Price and Low Price

are positively correlated with Close Price. For the three macroeconomic variables, Inflation rate has a positive correlation with stock price, while both Interest rate and Unemployment rate have a negative correlation with stock price, which is in line with the objective relationship between macroeconomics and stock market, i.e., changes in discounted cash flow due to changes in interest rates in macroeconomics have a negative relationship with the stock market. Unemployment will erode investor confidence, and the study finds that unemployment rate has little effect on Amazon and Netflix relative to the broad market index Nasdaq and the large manufacturing firms Tesla and Google, proving that it has less of an impact on the service sector.

5.2 Limitations

Firstly, the dataset of this study uses 36 months of data from 2020 to 2022, which is an insufficient sample of data and may result in the predictive model failing to capture long-term trends or cyclical behaviours of the broader stock indices and company-specific stocks. Second, the stock market is often affected by external shocks that cause volatility, such as financial news, geopolitical crises, or natural disasters, making it difficult to predict future stock prices using only past data. At the same time, this was the period of the Covid-19 global pandemic, and it is difficult to find its relationship with the stock market or specific stocks through quantitative research alone. Finally, macroeconomic indicators are often reflected in the stock market with a lag, complicating forecasting. Investor sentiment and behaviour play an important role in stock prices, but these factors are difficult to quantify for forecasting purposes.

5.3 Recommendations for the Future Research

Future research should use a longer and more flexible time horizon to capture price trends in the stock market, which can help researchers to find seasonal or trend changes by looking at broad market indices or specific stock prices over time. The next step is to extend the dataset features to capture market sentiment from social media, news etc. to predict stock movements. It is well known that social media has a huge influence, people collect and share business information through social media, and investor sentiment largely affects the stock price of a company or industry. Finally, with the

surge in computational power, better predictive power can be obtained using more advanced neural networks and LSTMs. They are able to extract features and handle unstructured data processing, thus capturing long-term dependencies in financial data.

Finally, the world of the stock market is complex, full of impossible changes and unpredictable factors, and forecasting is a challenging and desirable task when the ability to see trends is something that everyone wants to have. However, with today's information technology, from traditional statistical methods to deep learning, stock market prediction is no longer out of reach.

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Appendix

Appendix 1. Ethics Documents



Application 056304 **Self-declaration**

Section A: Applicant details
Date application started: Sun 16 July 2023 at 19:04
First name: Wenliang
Last name: Mo
Email: wmo3@sheffield.ac.uk
Programme name: Information Systems Management
Module name: INF6000 Dissertation
Last updated: 20/07/2023
Department: Management School
Applying as: Undergraduate / Postgraduate taught
Research project title: The Role of Social Media Sentiment in Stock Market Prediction
Similar applications: - not entered -

Section B: Basic information				
Supervisor				
<table><tr><th>Name</th><th>Email</th></tr><tr><td>Venet Osmani</td><td>v.osmani@sheffield.ac.uk</td></tr></table>	Name	Email	Venet Osmani	v.osmani@sheffield.ac.uk
Name	Email			
Venet Osmani	v.osmani@sheffield.ac.uk			
Proposed project duration				
Proposed start date: Sun 16 July 2023				
Proposed end date: Mon 28 August 2023				

Section C: Summary of research
1. Aims & Objectives

Aim:

The dissertation aims to explore the relationship between social media sentiment and stock market performance using quantitative methods. With the rapid development of social media, more and more studies on the use of social media data in financial decision-making contribute. This study hopes to help investors understand the role of social media in the stock market and predict stock market performance based on social media data and traditional financial data.

Research Objectives:

1. Review the literature on social media sentiment and stock market forecasting and find the research gap.
2. Data mining for social media data and making classification.
2. Analyse the relationship between social media sentiment and stock market volatility.
3. Use social media sentiment to predict the stock market and compare its accuracy.
4. Identify the limitations and challenges of this study.

Research Questions:

1. Is there a correlation between social media sentiment and stock realization.
2. Which algorithms can be applied to the analysis and prediction of social media sentiment on stocks?
3. How accurate are these algorithms?

The Research gap:

Numerous studies have been conducted to investigate the relationship between social media sentiment and stock market performance, with mixed results. Some studies have found a positive relationship between social media sentiment and stock prices, while others have found no significant relationship. In addition, previous studies have limited consideration of the influence of external factors on the analysis of emotional factors, which will have limitations on the research results. For example, differences in the emotional analysis of social media may be affected by regions, cultures, races, etc. If the problem is unified, it is difficult to obtain A comprehensive study. Therefore, I hope to verify the relationship between social media sentiment and stock market performance through this article. I will select or even select a few industries to study, hoping to draw universal conclusions.

2. Methodology

In the data collection part, I will download company historical stock data from some public websites, such as Yahoo Finance, etc. And collect social media sentiment data from a public database.

In the data analysis part, I will use NLP to analyze and classify social media data. Use hypothesis testing to study the relationship between social media sentiment and stock performance, and then use a simple statistical analysis method such as regression analysis and simple machine learning methods, such as decision tree and SVM to do stock market prediction and test their accuracy.

Section F: Supporting documentation

Additional Documentation

[Document 1126655 \(Version 1\)](#)
Dissertation Proposal

[All versions](#)

External Documentation

- not entered -

Section G: Declaration

Signed by:
Wenliang Mo
Date signed:
Thu 20 July 2023 at 15:09

Official notes

- not entered -

Appendix 2. Ethics Approval Letter



Downloaded: 24/07/2023
Approved: 24/07/2023

Wenliang Mo
Registration number: 220249494
Information School
Programme: Information Systems Management

Dear Wenliang

PROJECT TITLE: The Role of Social Media Sentiment in Stock Market Prediction

APPLICATION: Reference Number 056304

This letter confirms that you have signed a University Research Ethics Committee-approved self-declaration to confirm that your research will involve only existing research, clinical or other data that has been robustly anonymised. You have judged it to be unlikely that this project would cause offence to those who originally provided the data, should they become aware of it.

As such, on behalf of the University Research Ethics Committee, I can confirm that your project can go ahead on the basis of this self-declaration.

If during the course of the project you need to [deviate significantly from the above-approved documentation](#) please inform me since full ethical review may be required.

Yours sincerely

Claire Du Puget
Departmental Ethics Administrator

Appendix 3. R output of TSLA by Linear Regression

	Estimate	Standard Error	t value	Pr(> t)
(Intercept)	0.044	0.062	0.720	0.4793
Open	-0.647	0.130	-4.988	0.0001 ***
High	0.837	0.141	5.927	0.0000 ***
Low	0.646	0.168	3.843	0.0009 ***
Volume	0.076	0.067	1.137	0.2683
IR	-0.452	0.184	-2.456	0.0228 *
Inflation.rate	0.125	0.054	2.300	0.0318 *
UNRATE	-0.097	0.070	-1.389	0.1794

*Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05*

Residual standard error: 0.0432 on 21 degrees of freedom

Multiple R-squared: 0.9849, Adjusted R-squared: 0.9799

F-statistic: 195.6 on 21 and 7 DF, p-value: 0.0000

Appendix 4. R output of Amazon by Linear Regression

	Estimate	Standard Error	t value	Pr(> t)
(Intercept)	0.174	0.149	1.168	0.2561
Open	-0.457	0.178	-2.567	0.0180 *
High	0.430	0.229	1.879	0.0742 .
Low	0.785	0.178	4.402	0.0002 ***
Volume	0.035	0.094	0.368	0.7164
IR	-0.443	0.332	-1.332	0.1970
Inflation.rate	0.081	0.082	0.992	0.3324
UNRATE	-0.005	0.130	-0.036	0.9719

*Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05*

Residual standard error: 0.06636 on 21 degrees of freedom

Multiple R-squared: 0.9522, Adjusted R-squared: 0.9362

F-statistic: 59.75 on 21 and 7 DF, p-value: 0.0000

Appendix 5. R output of Netflix by Linear Regression

	Estimate	Standard Error	t value	Pr(> t)
(Intercept)	0.183	0.074	2.480	0.0217 *
Open	-0.564	0.220	-2.560	0.0183 *
High	0.945	0.280	3.372	0.0029 **
Low	0.339	0.187	1.817	0.0835 .
Volume	-0.293	0.103	-2.858	0.0094 **
IR	-0.037	0.144	-0.259	0.7985
Inflation.rate	0.024	0.045	0.528	0.6028
UNRATE	-0.000	0.064	-0.005	0.9957

Signif. codes: 0 <= '****' < 0.001 < '***' < 0.01 < '**' < 0.05

Residual standard error: 0.042 on 21 degrees of freedom

Multiple R-squared: 0.973, Adjusted R-squared: 0.9639

F-statistic: 107.9 on 21 and 7 DF, p-value: 0.0000

Appendix 6. R output of Google by Linear Regression

	Estimate	Standard Error	t value	Pr(> t)
(Intercept)	-0.015	0.066	-0.233	0.8179
Open	-0.759	0.144	-5.271	0.0000 ***
High	0.589	0.163	3.609	0.0016 **
Low	1.080	0.160	6.765	0.0000 ***
Volume	0.073	0.071	1.026	0.3167
IR	-0.069	0.144	-0.480	0.6360
Inflation.rate	0.112	0.076	1.482	0.1533
UNRATE	0.023	0.058	0.408	0.6875

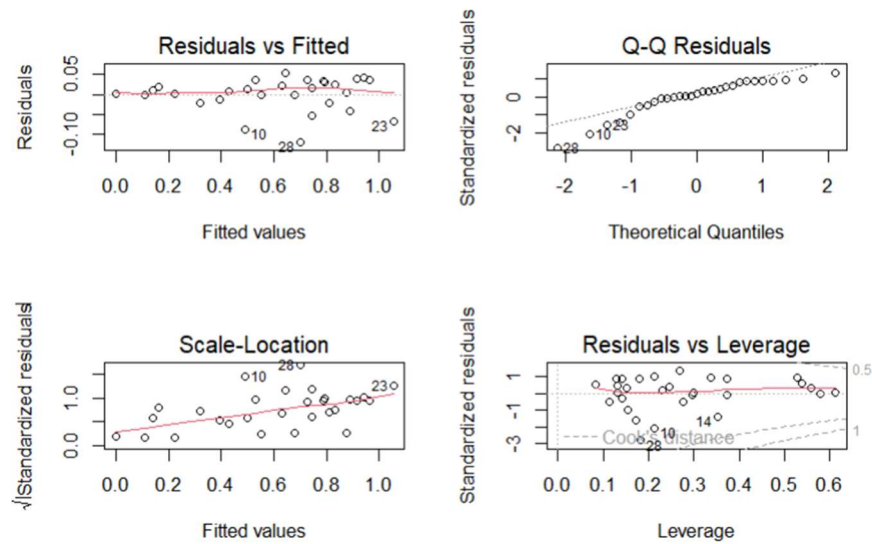
Signif. codes: 0 <= '****' < 0.001 < '***' < 0.01 < '**' < 0.05

Residual standard error: 0.03442 on 21 degrees of freedom

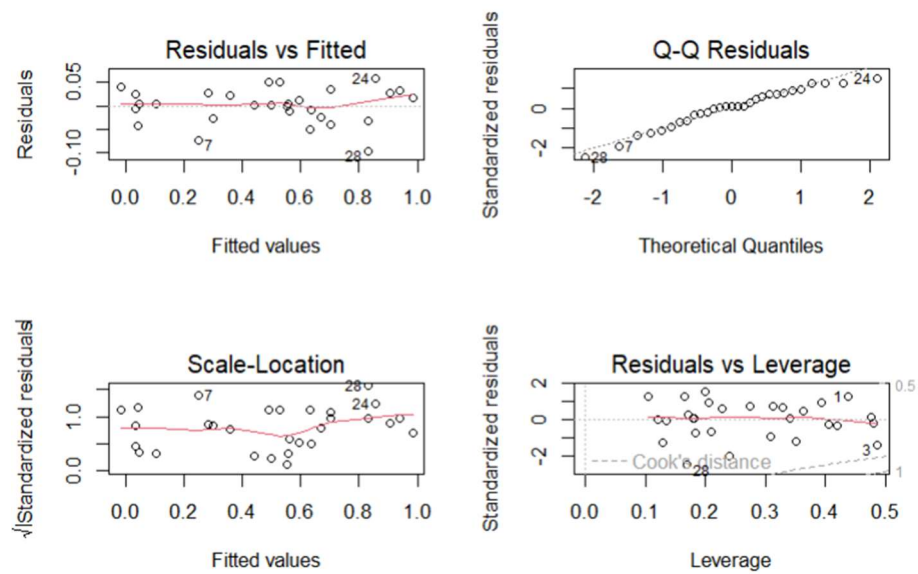
Multiple R-squared: 0.9918, Adjusted R-squared: 0.9891

F-statistic: 362.4 on 21 and 7 DF, p-value: 0.0000

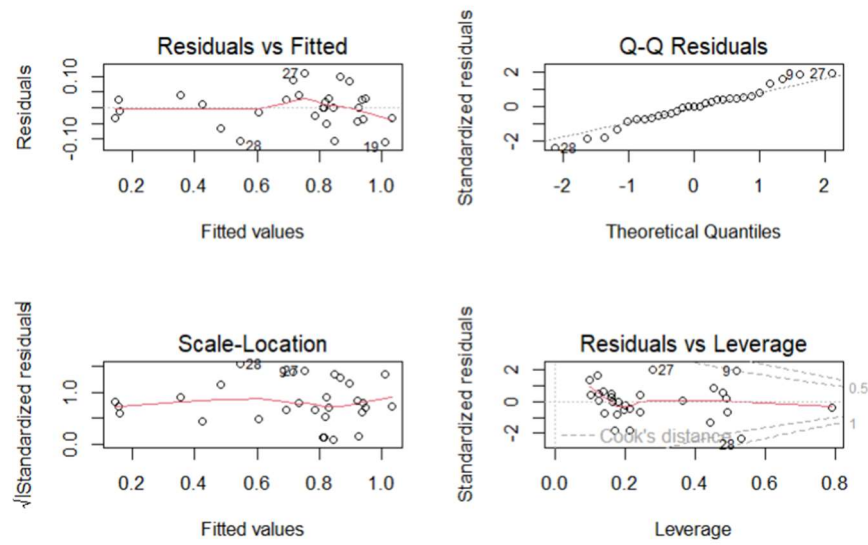
Appendix 7. Residuals of Nasdaq Composite by Linear Regression



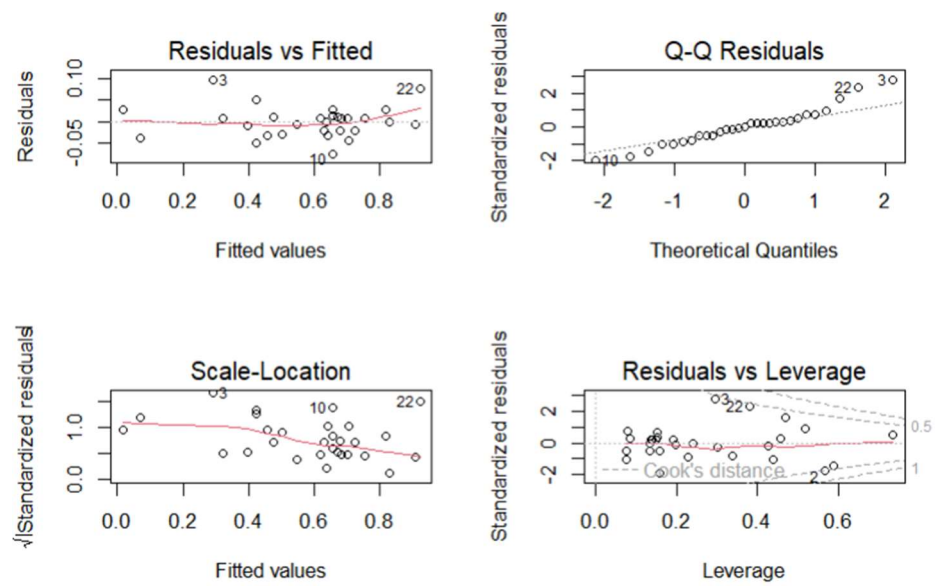
Appendix 8. Residuals of Tesla by Linear Regression



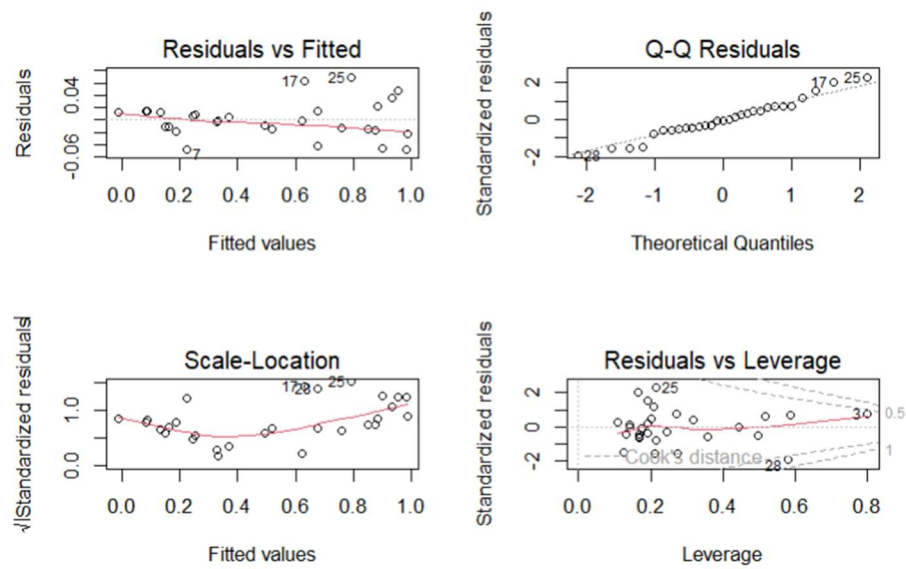
Appendix 9. Residuals of Amazon by Linear Regression



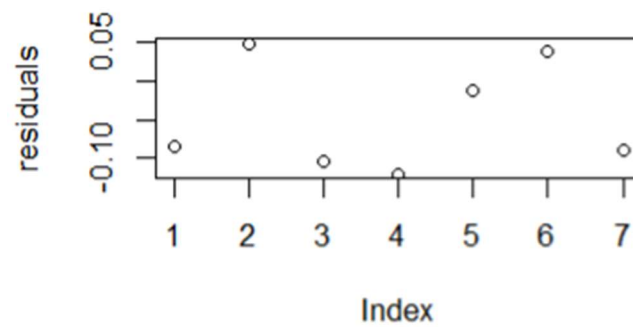
Appendix 10. Residuals of Netflix by Linear Regression



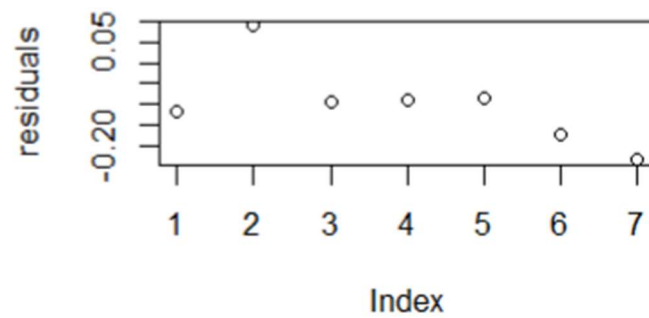
Appendix 11. Residuals of Google by Linear Regression



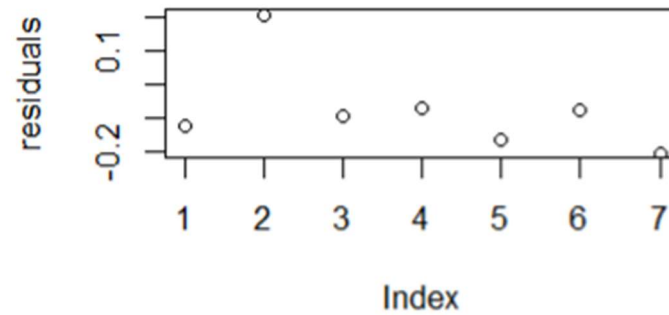
Appendix 12. Residuals of Nasdaq Composite by Random Forest



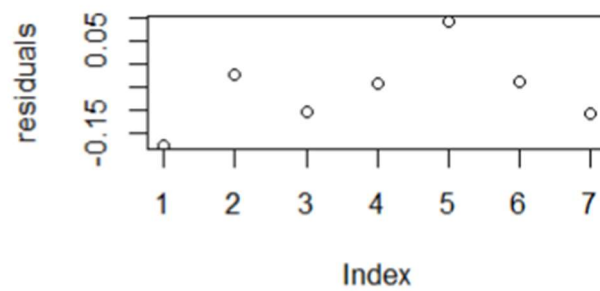
Appendix 13. Residuals of Tesla by Random Forest



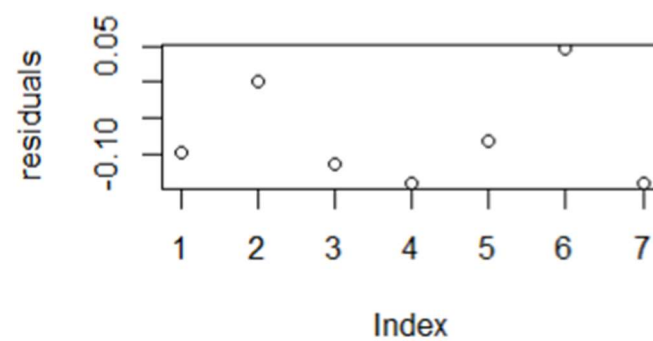
Appendix 14. Residuals of Amazon by Random Forest



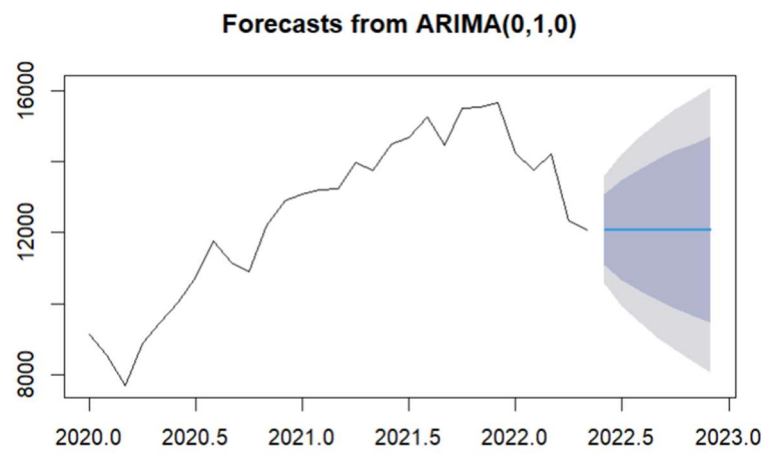
Appendix 15. Residuals of Netflix by Random Forest



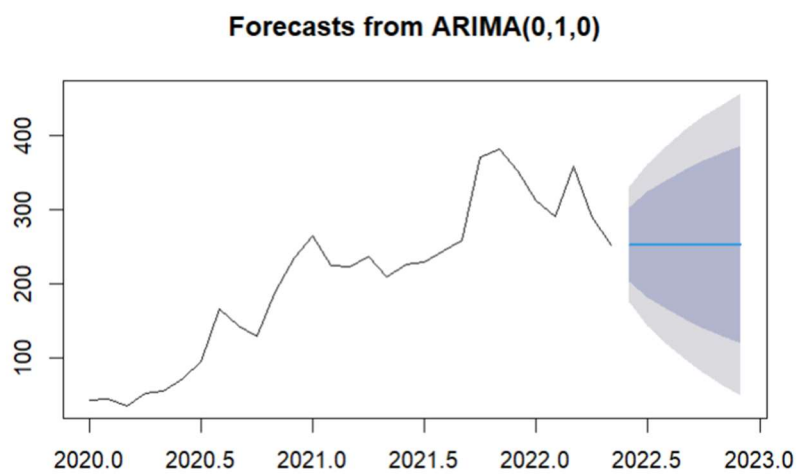
Appendix 16. Residuals of Google by Random Forest



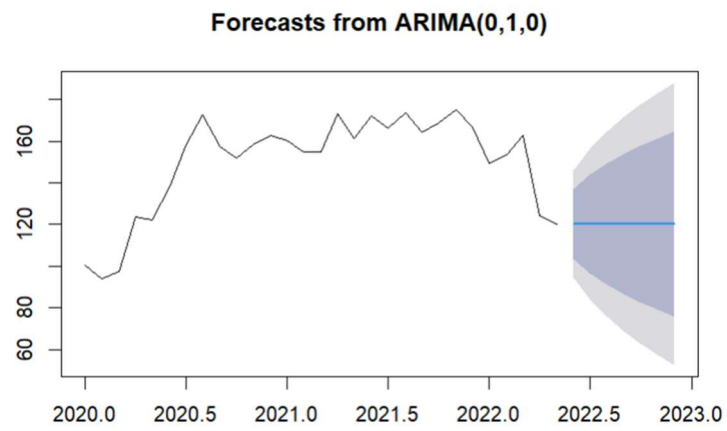
Appendix 17. Time Series Result of Nasdaq



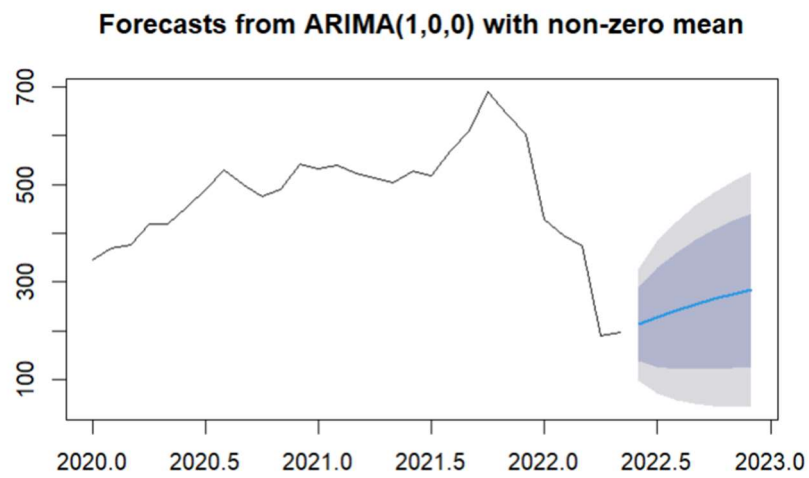
Appendix 18. Time Series Result of Tesla



Appendix 19. Time Series Result of Amazon



Appendix 20. Time Series Result of Netflix



Appendix 21. Time Series Result of Google

